장신우 논문

[답변] 11 월 1 일 수정 사항

1. 초록: 숫자를 빼고 여섯 문장으로 다시 써서 발견과 기여만 남겼습니다.

2. 서론 기여 문단: 가장 가까운 선행연구(Duarte & Eisenbach, 2021; Caccioli et al., 2024; Capponi & Jia, 2025; Lehar & Parlour, 2025)를 먼저 밝히고, 본 연구가 더하는 점을 이어서 쓰도록 고쳤습니다.

3. 해석 문장: 인과처럼 읽히는 표현을 연관성 표현으로 바꿨습니다(5 장 표 3 설명, 7 장 표 8·표 9 설명, 결론).

4. 본문 수치: 7 장 표 9 설명에 있던 p 값을 빼고 유의수준만 적었고, 쉼표가 빠진 문장을 고쳤습니다.

5. 표 주석: 모든 표 주석을 네 문장 이내로 줄였고, 첫 문장에 관측 단위를 적었으며, 괄호 안 t 값의 표준오차를 완전한 문장으로 설명했습니다(표 1, 3–10, A1, B1–B4, C1).

6. 용어: chain-date, wild-cluster, market- and date-fixed effects 같은 하이픈 표현을 풀어 썼습니다.

7. 선언문: 저자 기여(CRediT), 이해상충, 생성형 AI 사용 선언을 데이터 이용 가능성 뒤에 추가했습니다.

8. 참고문헌: 게재 예정 논문은 In-press 로 표기했습니다.

9. 서식: 표 머리글 반복과 '다음 단락과 함께' 설정을 모두 해제했습니다.

표와 노트 및 형식 및 해석 관련해서 기존에 내가 손한, 이재호 논문에 대해서 준 comments 복기하여 한번더 간결하고 세련되게 수정하십시요.

[답변] 손한 논문에 주신 지적을 이 원고에 같은 기준으로 다시 적용했습니다. 표 안의 이탤릭을 모두 풀었고, 모든 표를 한 가지 양식으로 맞췄습니다. 회귀표는 열 머리글을 열마다 나누고 계수 구역과 아래 구역 사이에 선을 넣었습니다. 주에서는 표에 나오는 항목 이름을 빠짐없이 이탤릭으로 맞췄고, 별표 설명이 표에 실제로 나온 수준과 같은지 표마다 확인했습니다. 각주는 12 포인트, 줄간격 1.0 으로 고치고 첫 줄 들여쓰기를 없앴습니다. 본문은 긴 문장을 짧게 나누고 쉬운 단어로 다시 썼으며, 용어를 standalone capacity, joint cash bound, duplication rate 로 통일했습니다.

또한, 아래 Check Lists 에 대해서 상세하게 답변하고(AI 시키지 말고, 직접하시오), 수정하고, 필요시 반박하시오.

[답변] 항목마다 아래에 답을 적었습니다. 6 번, 7 번, 10 번은 자료를 다시 분석해 확인한 결과를 넣었습니다.

[중요] 표가 20 개 가까이 되고, 페이지도 너무 깁니다. 중복되는 내용 제거하고, 간결하게 더 압축해야함!

[답변] 본문 표를 19 개에서 11 개로, 그림을 6 개에서 5 개로 줄였습니다. 이전 번호 기준으로 표 4 와 5, 표 11 과 12, 표 15 와 17 은 각각 한 표의 패널로 합쳤습니다. 표 2, 13, 14, 16, 18 과 표 1 의 패널 B·C, 그림 3 은 핵심 수치만 본문 문장으로 옮기고 삭제했습니다. 겹치는 설명 문장을 지우고 초록, 서론의 결과 요약, 결론을 줄여 원고가 약 35 쪽에서 31 쪽이 되었습니다.

Table 10 의 overlapping events 와 Table 19 의 "실제 withdrawal failure" 분류는 분석 또는 주장 범위를 한 번 더 정리해야 함.

[답변] 두 표 모두 다시 분석해 주장 범위를 좁혔습니다. 내용은 6 번과 7 번에 적었습니다.

Check Lists

1. duplication rate 를 아직 한 곳에서 잘못 해석합니다

Section 4 에서 현재:

"On the median day 57.86 percent of summed standalone capacity is cash that at least one other vault also counts."

라고 씁니다.

이것은 정확하지 않습니다.

\[D=(S-J)/S \]

는 **summed standalone capacity 가 joint cash bound 를 초과하는 비율**이지, "다른 vault 도 함께 세고 있는 cash 의 비율" 과 동일하지 않습니다.

예를 들어 두 vault 가 같은 $100 를 각각 capacity 로 세면 \(S=200\), \(J=100\), \(D=50\%\)입니다. 그러나 두 vault 의 $200 capacity 전체가 같은 $100 에 의존합니다.

따라서 반드시 다음처럼 바꾸십시오.

On the median day, the excess of summed standalone capacity over the joint cash bound equals 57.86 percent of summed standalone capacity.

같은 이유로 조금 뒤의

"close to two thirds of standalone capacity rests on cash that another vault also counts"

도

the excess over the joint cash bound amounts to close to two thirds of summed standalone capacity

로 바꾸는 것이 좋습니다.

이 문제는 **개념 정의와 바로 연결되므로 필수 수정**입니다.

[답변] 지적이 맞습니다. 첫 문장은 적어 주신 대로 고쳤습니다. 둘째 문장은 표 1 의 패널 C 를 본문으로 옮기면서 삭제했고, 그 자리에는 9 월 15 개 수집일의 범위 (61.62~63.74%)만 적었습니다. 바로 뒤의 Spark 와 주 시장을 제외한 경우 문장도 초과분의 비율이라는 뜻으로 고쳤고, 하이라이트 둘째 줄은 금고들이 함께 인출할 수

있는 금액이 합계의 절반에 못 미친다고 썼습니다.

2.Highlights 와 Abstract 첫 문장도 조금 강합니다

현재 Highlight:

"Vaults that lend into the same markets report the same idle cash."

그리고 Abstract 첫 문장도 거의 같은 표현입니다.

엄밀히는 vault 가 "same idle cash 자체를 report"하는 것이 아니라, **각 standalone withdrawal capacity 가 동일 market cash 를 이용할 수 있습니다.**

예를 들어 어떤 vault 의 position 이 market cash 보다 작으면 그 vault 가 market idle cash 전액을 capacity 로 세는 것도 아닙니다.

더 정확하고 여전히 강한 Highlight 는:

Vault-level withdrawal capacities can draw on the same market cash.

Abstract:

Decentralized lending vaults each report how much their depositors could withdraw, but the standalone capacities of vaults using the same markets can draw on the same idle cash.

이 표현이 오히려 referee 에게 더 신뢰감을 줍니다.

[답변] 하이라이트 첫 줄과 초록 첫 문장을 적어 주신 표현으로 바꿨습니다. 하이라이트 둘째, 셋째 줄도 쉬운 표현으로 다시 썼습니다.

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3. Introduction 의 "every market that several vaults share"는 과도합니다

현재 Introduction:

"The gap is present throughout the sample, in every market that several vaults share and on every chain …"

공유 자체만으로 duplication 이 생기는 것은 아닙니다. 이미 뒤에서 정확하게

"Shared markets create scope for duplication once the participating vaults' combined positions exceed the market's cash"

라고 고쳤습니다.

Replication output 를 보면 실제로 여러 vault 가 사용하더라도 combined positions 가 cash 를 넘지 않는 경우 duplication 이 0 이 될 수 있습니다.

따라서 Introduction 도 같은 논리를 따라야 합니다.

추천: (**그대로 복붙하지 말고, 참고만 하여 수정하시오**)

The gap persists throughout the ten-vault sample and appears across all four chains; at the market level, it arises when the combined positions of vaults sharing a market exceed the market's available cash.

[답변] 서론 문장을 고쳤습니다. 차이가 열 개 금고 표본의 전 기간과 네 체인에서 나타난다고 쓰고, 시장 단위에서는 시장을 함께 쓰는 금고들의 포지션 합이 시장 현금을 넘을 때 생긴다는 문장을 따로 두었습니다.

4. Section 2 에 아직 "order of arrival"가 남아 있습니다

Section 2 에는 여전히:

"The mechanism is a common cash pool served in order of arrival."

이 남아 있습니다. 바로 다음 문장에서는 오히려 정확하게 "recorded on chain, which reflects block construction rather than the order in which requests were sent"라고 씁니다.

두 문장이 서로 충돌합니다.

그냥:

The mechanism is a common cash pool served sequentially.

또는

The mechanism is a common cash pool from which withdrawals are executed sequentially in recorded on-chain order.

로 바꾸면 끝입니다.

[답변] 첫 문장을 인출이 순차적으로 집행된다는 표현으로 바꿨습니다. 블록에 기록된 순서라는 설명은 뒤 문장 한 곳에만 두어 두 문장이 충돌하지 않게 했습니다.

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5. Table 5 의 해석에는 실제 계수와 반대 방향인 문장이

본문은:

"Low idle cash raises the double count mainly where several vaults are counting it…"

라고 씁니다.

그런데 Table 5 column (3)의 Cash coefficient 는 **+0.030**, t=3.08 로 유의합니다. 즉 Cash 가 log idle cash 라면 **cash 가 더 많을수록 MDup 가 높다는 조건부 계수**입니다.

따라서 저 문장은 표에서 나오지 않습니다.

이 문장은 삭제하고 다음 정도로 쓰는 것이 정확합니다. (**아래 그대로 복붙하지말 것!**)

The utilization association is strongest in markets shared by more vaults. The coefficient on cash itself is not stable across specifications, so the result is better interpreted as an interaction between market utilization and the number of vaults drawing on the market rather than as a direct effect of low cash balances.

이것은 **반드시 수정해야 합니다.**

[답변] 그 문장을 삭제했습니다. 가동률 계수가 금고가 많은 시장에서 더 크다는 점과, 현금 계수는 열마다 부호와 유의성이 달라 낮은 현금의 직접 효과로 읽을 수 없다는 점을 표의 계수대로 다시 썼습니다.

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6. Table 10: 현재 이 논문의 가장 큰 econometric weakness 입니다

72 시간 이내 다음 event 가 있는 사건 298 건, 1 주 내 560 건이며 isolated-event 재추정은 아직 하지 못했다고 답했습니다. Shared_liquidity_review_2026100…

현재 원고도 이를 감추지 않고 명시합니다.

그리고 Notes 에도 event clustering 이 **successive event windows 간 overlap 을 처리하지 않는다**고 정확히 씁니다.

이것은 투명한 서술이지만, **저널 referee 가 그대로 받아들일 가능성은 낮습니다.**

834 건 중 560 건은 168 시간 내 같은 market 의 다음 qualifying borrow 가 있습니다. 그렇다면 +72h 및 +168h coefficient 를 "첫 borrow 의 지속 효과"라고 읽기 어렵습니다.

다음 qualifying borrow 직전에서 event window 를 censor 하고 재추정. (이를 꼭 하세요)

또는 최소한,

clean-event robustness: event 전후 ±72h 또는 ±168h 내 다른 qualifying borrow 가 없는 isolated events 만 재추정.

결과가 비슷하면 Appendix 한 표로 끝낼 수 있습니다.

[답변] 다음 기준 초과 차입 직전에서 사건 창을 자르고 다시 추정해 표 8(이전 표 10)의 (2)열에 넣었습니다. 사건 블록 계수는 그대로입니다(중복률 0.016, 공동 현금 한도 -0.225). 그 뒤로는 중복률이 1 시간, 공동 현금 한도가 6 시간까지만 10% 수준에서 같은 방향으로 유의합니다. 24 시간과 72 시간에는 둘 다 유의하지 않고, 168 시간에는 남은 130 건의 중복률이 오히려 낮게 나옵니다(10% 수준). 그래서 (1)열의 72 시간, 168 시간 계수는 첫 차입의 지속 효과가 아니라 뒤따른 차입이 섞인 경로라고 본문을 고쳤습니다. 전후 168 시간 안에 다른 기준 초과 차입이 없는 80 건만으로도 추정해 본문에 적었습니다(사건 블록 0.022, t = 1.45 와 -1.059, t = -1.97). 지난번에 적은 298 건, 560 건은 표본에 남긴 사건끼리만 센 값이었습니다. 표본에서 뺀 24 시간 내 차입까지 세면 72 시간 안 547 건, 168 시간 안 668 건이어서 이 숫자로 바로잡았습니다.

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7. Table 19: "failed exit"라는 용어는 아직 한 단계 강합니다

현재 Table 19 는 훨씬 좋아졌습니다. owner 와 caller 가 다른 사건 5 건도 공개했고, enclosing transaction 이 completed 인지, UserOperation failure 인지도 구분하며, 다음 withdrawal 이 initial rejected amount 전체 회수를 의미하지 않는다는 점도 명시합니다.

다만 한 가지 남았습니다.

Filter 가 다음 조건입니다.

request does not exceed the **owner's claim at the preceding block**

입니다.

그런데 liquidity error 가 contract execution 중 owner balance/allowance 검증보다 앞서 발생할 수 있다면, **liquidity revert 가 있었다는 것만으로 그 call 이 다른 모든 조건을 충족한 economically valid withdrawal 였다고 확정할 수 없습니다.**

특히 238 건은 enclosing transaction 자체가 성공했고, 그 가운데 185 건만 failed smart-contract user operation 으로 식별됩니다. 즉 나머지 경우에는 caller/router 가 revert 를

catch 했을 가능성도 있습니다.

따라서 두 방법 중 하나가 필요합니다.

가장 좋은 방법

call 직전 exact state 에서:

- owner share/claim
- allowance when caller ≠ owner
- same-transaction fallback withdrawal
- eventual same-transaction success

를 확인합니다.

그리고 Section 6 에서 "realized withdrawal failures"보다는:

observed liquidity-error calls

을 쓰는 편이 안전합니다.

[답변] 표 11(이전 표 19)과 본문의 용어를 유동성 오류 호출로 바꾸고 주장 범위를 줄였습니다. 금고 계약에서는 유동성 확인이 지분 확인과 허용 한도 확인보다 먼저 실행되므로, 오류만으로는 다른 조건을 모두 충족한 인출이었다고 할 수 없다고 본문에 밝혔습니다. 252 건을 다시 확인하니 같은 거래 안에서 같은 보유자의 인출이 완료된 경우는 한 건도 없습니다. 거래가 완료된 238 건은 스마트 계정의 실패한 작업 185 건과 호출한 계약이 오류 뒤에도 계속 실행된 53 건이고, 거래 전체가 되돌려진 14 건에 호출자와 보유자가 다른 5 건이 모두 들어 있습니다. 이 5 건의 허용 한도는 관측하지 못했고, 보유자 청구액은 호출 직전이 아니라 직전 블록 기준이라는 점을 그대로 적었습니다.

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8. Table 19 와 joint cash bound 의 관계도 Conclusion 에서 조금 과하게 연결됩니다

Conclusion:

"Recorded refusals are concentrated in one small vault … so the joint bound is a constraint that large simultaneous requests would meet rather than one that has been reached in realized flows."

Table 19 의 liquidity-error call 은 **개별 vault 의 executable liquidity shortfall** 을 보여줍니다. 이것이 cross-vault joint bound 가 실제로 exhausted 되었거나 exhausted 되지 않았음을 직접 식별하지는 않습니다.

따라서:

Recorded liquidity-error calls are concentrated in one small vault, with one large event elsewhere. These observations show that local cash constraints can bind in realized transactions, while the joint-budget analysis addresses a different question: whether large simultaneous requests that pass individual checks could all be supported by the cash available at a common state.

이렇게 분리하는 것이 가장 좋습니다.

이 문장을 고치면 논리 구조가 오히려 더 강해집니다.

[답변] 결론의 그 문장을 나눠 썼습니다. 유동성 오류 호출은 개별 금고의 현금 제약이 실제 거래에서 걸릴 수 있음을 보여 주고, 공동 한도 분석은 개별 확인을 통과한 큰 요청들이 같은 시점의 현금으로 모두 지급될 수 있는지라는 다른 질문을 다룬다고 썼습니다. 6 장 끝에도 같은 구분을 넣었습니다.

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9. Appendix B 는 현재 Notes 와 본문 사이에 약간의 contradiction 이 있습니다

본문은:

"For each population … the two quantities are constructed exactly as for Base."

라고 합니다.

그런데 Notes 는 Euler 와 Yearn 에서는 **strategy 가 underlying protocol 에서 즉시 회수할 수 있는 자금을 포함하지 않고 idle cash 만 세므로 Morpho 보다 narrower measure** 라고 정확히 인정합니다. [wk]zhangr,Kang,ryu,sun(2610.05…

답변서에서도 "재측정하지 않았고 Morpho 보다 좁은 측정치라고 밝혔다"고 합니다.

그렇다면 constructed exactly as for Base 는 삭제해야 합니다.

추천:

For each protocol-chain population, we construct the closest contract-level analogue of standalone and joint immediately available cash. Because strategy-level recovery rules differ across protocols, the Euler and Yearn measures are narrower than the Morpho measure and should be interpreted as supporting evidence rather than directly comparable estimates.

[답변] 해당 표현을 삭제했습니다. 계약 수준에서 가장 가까운 대응 지표를 구성했다고 쓰고, Euler 와 Yearn 수치는 Morpho 보다 좁은 측정이어서 직접 비교할 추정치가 아니라 보조 증거라고 본문에 적었습니다.

---

10. V1.1 loss ledger

Replication Package 의 raw cross-chain JSON 을 실제 확인하면 **nonzero lostAssets 가 존재합니다.** 극히 사소한 1–2 micro-unit 만 있는 것이 아니라 일부 state 에는 material amount 도 있습니다.

따라서 작은 appendix robustness 로 처리하는 것이 좋습니다.

예를 들어:

Number of vault-states with lostAssets > 0

lostAssets / total assets

main Table 6 regression excluding affected vault-states

세 줄짜리 diagnostic 이면 충분합니다.

Main coefficient 가 그대로라면 오히려 referee 의 잠재 질문 하나를 선제적으로 없앱니다.

[답변] 확인해서 부록 B 에 넣었습니다. 손실 장부가 0 이 아닌 관측은 2,141 개 중 161 개(41 개 금고)이고, 1 USDC 를 넘는 것이 38 개, 금고 자산의 1%를 넘는 것이 31 개입니다. 이 161 개를 빼고 표 4(이전 표 6)의 (3)열을 다시 추정하면 Shared 0.551(t = 8.97), Util 0.055(t = 8.66)로 원래 값(0.566, 0.055)과 거의 같습니다.

---

11. 문법·표현

위치	현재	수정
Table 1 Notes	All ten vaults uses the full cohort	**The "All ten vaults" row uses the full cohort.**
Section 4	...rather than test behavior, large t-statistics are expected, and...	comma splice. 두 문장으로 분리
Appendix A	continuous linear-program checks agree to within USDC	**agree to within 1 USDC**
Table 17 first column	1,000,000	≥1,000,000
throughout	exit capacity, withdrawal capacity 혼용	**withdrawal capacity** 로 통일하는 편이 더 전문적
Table 1 title	Standalone cash, joint capacity and duplicated claims	**Standalone capacity, joint cash bound and duplicated capacity** 가 더 정확

Section 4 는 다음처럼 쓰면 더 깔끔합니다.

These regressions describe how the measured gap covaries with summaries of the same underlying positions and cash balances. They do not identify behavioral or causal responses. Large t-statistics are therefore unsurprising when variables share the same accounting inputs.

[답변] 여섯 곳 모두 고쳤습니다. 부록 A 의 허용 오차와 표 17 의 ≥ 기호는 수식 개체로 들어가 있어 보이지 않았던 것이어서 둘 다 일반 글자로 바꿨습니다. 부록 A 는 1 마이크로 USDC 이내로 일치한다고 썼고, 이전 표 17 은 표 10 의 패널 B 가 되었습니다. 4 장 문장은 적어 주신 대로 세 문장으로 나눴습니다.

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12. 경제학적 함의와 기여 (중요)

논문이 기계적으로 나열 하는 부분이 많습니다.

특히, same distinction, counterpart, parallels 식으로 기존 유명 연구와 계속 연결하는 표현을 줄여야 합니다.

현재 empirical result 가 독자적으로 의미가 있으므로 매 paragraph 마다 기존 이론의 analogue 라고 설명할 필요가 없습니다.

[답변] 결과를 기존 연구의 대응물로 설명하던 문장 여섯 곳을 삭제했습니다(4 장 두

곳, 5 장 두 곳, 6 장 두 곳). 삭제한 문장의 문헌은 서론에서 인용하고, 결과 문단에는 결과 자체의 의미만 남겼습니다. 2 장에서 순차 지급 제약을 정의하는 한 곳만 그대로 두었습니다.

---

13 Notes

아직 독립성을 조금 더 높일 수 있는 곳은 Figure 2 와 Figure 3 입니다. Figure 2 Notes 의 ten-vault cohort of Table 1, Figure 3 의 populations in Table 3 처럼 다른 표를 봐야 정확한 sample 을 아는 표현은 가능하면 직접 정의하십시오.

예:

The sample comprises the ten largest native-USDC Morpho vaults on Base by total assets on 16 January 2025.

Figure 3:

The figure covers all native-USDC Morpho V1 vaults with positive assets on Base and Ethereum at the synchronized 29 September 2026 states reported in the figure.

Notes 에서 결과의 의미를 장황하게 해석하지 않고 본문에 둔 현재 방식은 좋습니다.

[답변] 그림 1 과 그림 2 의 주에 표본을 직접 정의했습니다. 그림 3 은 표 2 와 내용이 겹쳐 삭제했습니다. 표 4 와 표 B2 의 주에서도 다른 표를 가리키던 표현을 줄였습니다.

Counting the same cash: shared market liquidity and withdrawal capacity across decentralized lending vaults

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Highlights

• Vault-level withdrawal capacities can draw on the same market cash.
• On the median day, the joint cash bound is under half of summed capacity.
• Large requests each vault could meet alone often cannot all be met at once.
• One deposit raises the withdrawal capacity of other vaults and holders.

Abstract

Individually valid liquidity figures need not be additive when financial intermediaries draw on common cash. Using synchronized contract states for the ten largest native-USDC lending vaults of the Morpho protocol on the Base network, we compare summed standalone withdrawal capacity with an upper bound on what the vaults could withdraw together. On the median day, the excess of summed capacity over this joint cash bound is more than half of summed capacity, and a similar gap appears on the other chains where several vaults share markets. Large simultaneous requests that every vault could meet on its own often exceed the joint cash bound, whereas small requests do not, and in a replay of recorded flows, large requests can take days to be paid in full. A single deposit into a shared market raises the withdrawal capacity of vaults and holders whose own positions do not change. The evidence suggests that standalone liquidity figures should be accompanied by a measure of the cash that vaults share.

Keywords: Decentralized lending; Financial intermediation; Joint liquidity capacity; Liquidity disclosure; Stablecoins; Withdrawal capacity

JEL classification: G01, G21, G23

1. Introduction

Research on liquidity transformation asks whether the claims that investment funds issue can all be honored at once. Holding a claim differs from having the cash to pay it (Jin et al., 2022; Ma et al., 2022), redemptions by some investors impose costs on those who remain (Falato et al., 2021), and funds that share a liquid buffer or hold each other pass stress to one another (Fricke & Wilke, 2023; Huang et al., 2025). For these funds, outsiders cannot see which liquidity is shared, so overlap has to be inferred from reported holdings. Decentralized lending vaults pool stablecoin deposits, lend them into shared markets, and each report how much cash their depositors could withdraw. For these vaults, every position, every market's idle cash, and every withdrawal rule is recorded on chain at the same block. We use these records to compute an upper bound on what a group of vaults could withdraw together and compare it with the figures the vaults report one by one.

Suppose two vaults have each supplied 100 to a market that holds 100 of idle cash. Each vault can withdraw 100, so the two figures add up to 200. Together, the vaults can withdraw only 100. Each figure is correct for its own vault, but whichever vault withdraws second finds the cash gone. We ask how large this gap is across real vaults, how long it lasts, and at which request sizes it matters.

We study the ten native-USDC vaults on the Morpho protocol on the Base network with the most assets on the first day of the sample and follow them for fourteen months. We then extend the measurement to every native-USDC vault of the same contract version on Ethereum, Base,

Arbitrum and Polygon. We call the amount a vault could withdraw alone its standalone capacity. We call the upper bound on what the vaults could withdraw together the joint cash bound. The excess of summed standalone capacity over the joint cash bound, as a share of that sum, is the duplication rate. On the median day, the excess of summed standalone capacity over the joint cash bound is 57.86 percent of summed standalone capacity. The gap persists throughout the sample and appears on all four chains. It arises in a market when the combined positions of the vaults that share it exceed the market's cash. The gap matters for large withdrawals. When every vault is asked for a quarter of its assets, the requests together exceed the joint cash bound on 62 of the 81 dates on which each vault could pay its own request. In a replay that holds other participants' recorded inflows fixed and pays the requests through successive partial withdrawals, the median time to full payment is 128 hours. Smaller requests are almost never affected. A single block shows the dependence directly: one holder's deposit raises the withdrawal capacity of two other vaults and of a second holder whose positions do not change. Across the sample, a vault's capacity rises when others add cash to its markets and falls when borrowers take cash out. A large borrow raises the market's duplication rate and lowers its joint cash bound, more so where more vaults share the market. Withdrawal calls that reverted with the vault's liquidity error are concentrated in the smallest vault, and in the November 2025 stress episode the joint cash bound fell sharply on Ethereum and Arbitrum.

The closest studies examine resources that several financial intermediaries share. Duarte and Eisenbach (2021) and Caccioli et al. (2024) show how overlapping holdings spread fire-sale losses across banks and non-banks, and Capponi and Jia (2025) and Lehar and Parlour (2025) show how pooled liquidity on decentralized exchanges becomes a common resource among its providers. In the fire-sale studies, the channel runs through prices, because one institution's sale lowers the value of what the others hold. This study measures a channel that runs through quantities. When several intermediaries draw on one pool of cash that is paid out one request at a time, one withdrawal lowers the cash the others can obtain, while the value of every claim is unchanged. The protocol describes the benefit of shared markets, since cash supplied by one vault becomes withdrawable by the others that use the same market.[1] We measure the other side of this benefit state by state with the complete position records of every participant, show at which request sizes it changes the assessment of simultaneous withdrawals, and separate an aggregate cash bound from payment under the vaults' withdrawal queues. The study also adds to research on decentralized lending, which documents who uses these protocols (Cornelli et al., 2025; Saengchote, 2023), how algorithmic rates clear a shared deposit pool (Son & Ryu, 2024), and how leverage and liquidation fall on pool users (Aspris & Svec, 2026). We show that a common-resource problem exists inside lending vaults and that figures reported vault by vault do not reveal it. Finally, research on stablecoins explains what keeps the token at par (Gorton et al., 2026; Lyons & Viswanath-Natraj, 2023), how reserves tie the token to short-term funding markets (Barthélemy et al., 2026), why issuers face runs (Ma et al., 2026), how stablecoins compare with money market funds (Oefele et al., 2024), and how they carry risk between crypto

[1] Morpho, Understanding Morpho Vaults: Aggregating & Amplifying Liquidity For Lenders, 23 February 2024, https://morpho.org/blog/understanding-morpho-vaults-aggregating-amplifying-liquidity-for-lenders. The post illustrates how supply by one vault raises the liquidity available to lenders in another vault that uses the same market.

and dollar markets (Huang et al., 2026). A stablecoin that trades at par does not guarantee that a vault share can be exchanged for it on demand, which matters because holdings are concentrated (Sapkota, 2025) and a large block can be worth less than its market value when sold (Garratt & van Oordt, 2026).

The remainder of this study is organized as follows. Section 2 defines the measures, Section 3 describes the data, Section 4 sets out the empirical approach, Sections 5 to 7 report the results, and Section 8 concludes.

2. Individual and Joint Liquidity Capacity

The setting has three layers: holders, vaults and markets. A holder owns shares of a vault. A vault lends the deposited USDC into several lending markets. Each market keeps the part that is not lent out as idle cash. The cash a holder can withdraw depends on the idle cash of the markets the vault uses and on the order in which the contract draws on them.[2] Each market serves several vaults, and withdrawals are executed one after another in the order recorded on chain. Public records make these positions visible, but they do not make owning a claim the same as being paid (Hautsch et al., 2024), nor do they disperse funding, risk and control (Chiu et al., 2022; Makarov & Schoar, 2022). The problem we study needs three conditions, and all three are visible in the records. Several vaults hold claims on one pool of cash. Their possible withdrawals together exceed that cash. Withdrawals are paid in recorded order rather than shared out in advance.

We define three quantities: standalone capacity, the joint cash bound and the duplication rate. For market *m*, let M_m be its idle cash and P_{vm} the amount vault *v* has supplied to it. The standalone capacity of vault *v* is the amount it could withdraw if no other vault withdrew,

$$S_v = \sum_m min\left(P_{vm}, M_m\right), \tag{1}$$

where the sum runs over the markets the vault uses and $min\left(P_{vm}, M_m\right)$ is the smaller of the vault's position and the market's idle cash. The vault cannot withdraw more than it supplied or more than the market holds. Adding the standalone capacities of a set *V* of vaults gives summed standalone capacity $S = \sum_{v \in V} S_v$. The joint cash bound is an upper bound on what the selected vaults can withdraw together from the cash available at that block,

$$J = \sum_m min\left(M_m, \sum_{v \in V} P_{vm}\right), \tag{2}$$

[2] Morpho Labs, MetaMorpho vault contract source, commit f308ddcd, https://github.com/morpho-org/metamorpho/blob/f308ddcdd3c1bb197e308a479cfedbd0a4b61f57/src/MetaMorpho.sol, and Morpho Blue market contract source, https://github.com/morpho-org/morpho-blue, together with the on-chain market-creation events; archived snapshots are in the replication files.

where each market's cash is counted once and compared with the combined positions of the selected vaults. The difference S-J is never negative, and the duplication rate is $(S-J)/S$ for $S>0$. For a single vault, $S=J$. Both quantities are calculated at each date before medians are taken. The bound is an upper limit, because each vault's contract draws on its markets in a fixed order, which can make the amount the vaults can actually withdraw together smaller than J.

The mechanism is a common cash pool from which withdrawals are paid one at a time. Consider one market with idle cash M that has received supply P_v from each of n vaults, with $\sum_v P_v > M$. Withdrawals are executed in the order recorded on chain. Each is paid in full if the cash is there at that moment and refused otherwise. This is the first-come, first-served rule of a bank run, applied to a contract. Let $X_{-v} \geq 0$ be the cash that other vaults withdraw before the request of vault v arrives. What vault v can obtain is

$$W_v = min\left(P_v, \left(M - X_{-v}\right)^{+}\right), \tag{3}$$

where $(x)^{+} = max(x, 0)$. Standalone capacity $min(P_v, M)$ is measured before any other vault withdraws, so it does not set that cash aside for the vault. The amount the vault can obtain even if every other selected vault withdraws first is its guaranteed capacity, $G_v = min\left\{P_v, \left(M - \sum_{u \neq v} P_u\right)^{+}\right\}$. The difference $min(P_v, M) - G_v$ is the cash that other selected vaults could take first. It is measured at one date, before later inflows.

Suppose every selected vault's request is within its own standalone capacity, but the requests together exceed the joint cash bound. Then the requests cannot all be paid from the cash available at that date, because any set of payments must respect each market's cash and each vault's position, and adding up these limits gives the joint cash bound in equation (2). No price enters these expressions. A withdrawal reduces the cash that others can obtain, while the value of every claim stays the same. The same accounting applies to holders inside one vault. A deposit d raises the capacity of holder j, whose claim Q_j is unchanged, by $min\left\{d, \left(Q_j - M\right)^{+}\right\}$. The gain is zero for a holder whose claim is already covered and at most the shortfall otherwise. Section 7 examines both the vault and the holder level.

In the data, we measure capacity as follows. For vault v at block t, Q_{vt} is the contract's total asset value and W_{vt} is the cash the vault could withdraw, rebuilt from its withdrawal queue. For each market in the queue, the available amount is the smallest of the vault's accrued position, the market's supply less its borrowing, and the USDC balance of the Morpho contract, and W_{vt} is the sum of these amounts. This calculation does not depend on who the holder is. For a holder o, the contract function maxWithdraw(o) caps the amount at the holder's own claim, and adding up these values across holders would count the same cash several times. A request for a fraction f of the vault's assets exceeds capacity when $fQ_{vt} > W_{vt}$, for f of 0.05, 0.10 and 0.25. Standalone coverage is S_{vt}/Q_{vt}.

The request sizes are standard amounts that we compare with past contract states. In the vault

sample, each amount is compared with the rebuilt capacity. In the holder tests, a request of fixed size is run against the archived state without sending a transaction. Completed withdrawals are observed separately in the transaction records. Capacity computed from positions and capacity rebuilt from the withdrawal queue agree exactly in every vault-state.

3. Data and Validation

The main sample is the ten largest native-USDC vaults on Morpho on Base by total assets on 16 January 2025. They are chosen from the vaults that existed before that date, so no vault is chosen because of its later overlap with others. They include Spark, Moonwell Flagship, Gauntlet Core, Gauntlet Prime and Extrafi and hold positions in 33 markets. The full sample uses 439 daily closing blocks from 16 January 2025 to 30 March 2026. The case window covers the 319 days to 30 November 2025. For September 2026, we collect the protocol's public interface on 15 dates and rebuild the same quantities from the chain at one block for each date.

At every daily block, we read each vault's withdrawal queue, supply positions and assets, and each market's balances and interest-rate inputs. The positions with accrued interest add up to the contract's totalAssets once the loss ledger of the V1.1 contract is taken into account. Our capacity calculation reproduces the maxWithdraw amounts that the Spark contract returns for individual holders at the same blocks. The public interface updates each vault at a different block, so we record that block for every vault and compute the joint cash bound only from states read at one common block.

The full sample contains 4,390 vault-days. For the Spark case, we use the states of individual holders, 94 matched deposits, a ledger of 18,967 addresses and archived runs of a request for a quarter of the claim. For the wider comparison, we read queues, positions and cash for every native-USDC V1 vault with positive assets that the protocol's interface lists, and the assets we rebuild equal totalAssets exactly. Finally, we scan every call to the 162 Base vaults over the sample period to find withdrawal calls that failed with the vault's liquidity error (Section 6). Amounts are million USDC unless specified otherwise.

Table 1 defines the variables and reports their summary statistics. The daily duplication rate is high on average and never falls below a third. The joint cash bound is a small fraction of the ten vaults' assets. The median vault-day exposure is close to one, so on a typical day the other nine vaults could take almost all of a vault's standalone capacity first.

Table 1. Variable definitions and summary statistics
Panel A. Variable definitions

	Definition
Daily series	
Dup	Duplication rate, summed standalone capacity less the joint cash bound divided by summed standalone capacity
JA	Joint cash bound divided by the ten vaults' total assets
HHI	Herfindahl index of idle cash across the markets in which the sample vaults hold positions
Size	Log of the ten vaults' combined assets in USDC
Util	Position-weighted utilization of the vaults' markets, per ten percentage points

Flow	Largest vault's net inflow divided by the ten vaults' assets
Vault-days	
D	Competing-cash exposure, the share of the vault's standalone capacity that the other sample vaults could take first
Shared	Share of the vault's positions in markets that at least one other sample vault uses
Size	Log of the vault's assets in USDC
Markets	Number of markets in which the vault holds a position
Market-days	
MDup	Within-market duplication rate among the sample vaults that use the market
Vaults	Number of sample vaults with a position in the market
Util	Market utilization, borrowing divided by supply, per ten percentage points
Cash	Log of the market's idle cash in USDC
Top	Largest single vault position divided by market supply
Four chains	
D	Competing-cash exposure measured against all other vaults on the vault's chain
Shared	Share of the vault's positions in markets in which another vault holds a position
Markets	Number of markets in which the vault holds a position
Size	Log of the vault's assets in USDC
Util	Position-weighted utilization of the vault's markets, per ten percentage points
Top	Largest market position divided by the vault's total positions
Daily changes	
ΔS	Change in the vault's standalone capacity over the day, million USDC
Own	Vault's own net supply into the sample markets over the day, million USDC
Peer	Net supply of the other nine sample vaults into the markets in which the vault holds a position at the opening block, million USDC
Outside	Net supply of all other suppliers into those markets, million USDC
Borrow	Net borrowing from those markets, million USDC
Shared	Opening share of the vault's positions in markets that another sample vault uses
Borrow events	
MDup	Within-market duplication rate at the event block
J	Market's joint cash bound at the event block, million USDC
Vaults	Number of sample vaults sharing the market at the block before the borrow

Panel B. Summary statistics

	Obs	Mean	SD	Min	P25	Med	P75	Max

Daily series								
Dup	439	0.584	0.071	0.347	0.540	0.579	0.644	0.752
JA	439	0.154	0.060	0.052	0.111	0.131	0.179	0.429
HHI	439	0.837	0.165	0.219	0.803	0.906	0.944	0.985
Size	439	19.92	0.821	18.13	19.09	20.31	20.60	20.84
Util	439	8.783	0.393	6.389	8.736	8.957	9.000	9.506
Flow	439	0.004	0.026	-0.228	-0.003	0.000	0.009	0.150
Vault-days								
D	4,387	0.779	0.377	0.000	0.797	0.998	1.000	1.000
Shared	4,387	0.756	0.383	0.000	0.615	1.000	1.000	1.000
Size	4,387	15.03	2.957	7.951	13.64	15.26	16.98	20.43
Markets	4,387	4.19	2.59	1	2	4	5	12
Market-days								
MDup	3,167	0.382	0.264	0.000	0.051	0.500	0.604	0.833
Vaults	3,167	3.84	2.08	2	2	3	5	9
Util	3,167	8.417	1.679	0.000	8.564	8.994	9.025	10.00
Cash	3,167	12.18	3.078	-10.01	10.09	11.78	13.35	19.62
Top	3,167	0.527	0.297	0.000	0.293	0.521	0.789	1.000
Four chains								
D	2,141	0.701	0.442	0.000	0.001	1.000	1.000	1.000
Shared	2,141	0.827	0.365	0.000	1.000	1.000	1.000	1.000
Markets	2,141	2.60	2.79	1	1	2	3	24
Size	2,141	4.896	8.484	-13.82	0.180	4.495	11.94	19.99
Util	2,141	7.044	3.341	0.000	7.394	8.665	9.018	10.00
Top	2,141	0.869	0.216	0.149	0.787	1.000	1.000	1.000
Daily changes								
ΔS	4,380	0.050	7.966	-111.6	-0.159	0.000	0.152	191.4
Own	4,380	0.121	4.847	-102.9	-0.099	0.000	0.117	102.4
Peer	4,380	0.796	12.85	-136.8	-0.866	0.253	4.048	117.9
Outside	4,380	1.046	12.76	-130.1	-0.901	0.000	1.546	179.0
Borrow	4,380	2.449	4.249	-46.52	0.027	2.046	4.444	30.81
Shared	4,380	0.755	0.383	0.000	0.609	1.000	1.000	1.000
Borrow events								
MDup	651	0.216	0.285	0.000	0.000	0.000	0.502	0.789
J	834	0.425	2.671	0.000	0.002	0.014	0.105	43.12
Vaults	834	2.06	1.53	1	1	1	3	8

Notes. Panel A defines the variables, and Panel B reports unweighted summary statistics for the daily series and the vault-day and market-day panels of the ten Base vaults from 16 January 2025 to 30 March 2026, the four-chain vault-states on seven dates from October 2025 to September 2026, the daily changes in capacity, and the large-borrow events. *Obs* is the number of observations, *Mean* the mean, *SD* the standard deviation, *Min* and *Max* the minimum and maximum, *P25* and *P75* the 25th and 75th percentiles, and *Med* the median. In the daily regressions, *HHI*, *Size*, *Util*, and *Flow* enter with their previous-day values. Shares are fractions, *Util* is in units of ten percentage points, *Size* and *Cash* are natural logarithms of

USDC amounts, and flows and *J* are in million USDC.

4. Empirical Approach

The analysis uses three sampling structures: daily closing states for the ten-vault panel, seven dated observations for the four-chain panel, and event-time states around large borrows. The aggregate daily regressions use lagged covariates, and the remaining specifications use the timing stated below. For the ten-vault sample, we first relate the daily duplication rate and the joint cash bound to the state of the markets on the previous day,

$$y_t = \alpha + \beta' X_{t-1} + \gamma_m + \delta_d + \varepsilon_t, \tag{4}$$

where y_t is the duplication rate Dup_t or the joint cash bound divided by the ten vaults' assets, JA_t. X_{t-1} collects the previous day's *HHI*, *Size* and *Util* for Dup_t, and the JA_t specification additionally includes the previous day's *Flow*. *HHI* is the concentration of idle cash across the vaults' markets, *Size* the log of their combined assets, *Util* the position-weighted utilization of their markets, and *Flow* the largest vault's net inflow divided by the ten vaults' assets. γ_m and δ_d are year-month and weekday effects. Standard errors are Newey-West with a bandwidth of seven days.

To ask which vaults and markets account for the gap, we measure for each vault its exposure to the other vaults, $D_{v,t} = [S_{v,t} - (J_t(V) - J_t(V_{-v}))]/S_{v,t}$, for $S_{v,t} > 0$, where the subscript -*v* removes vault *v* from the set. The numerator is the part of the vault's standalone capacity that the other nine vaults could take first. Market by market, the amount the vault keeps in that case is $min\{P_{v,m,t}, (M_{m,t} - \sum_{u \neq v} P_{u,m,t})^{+}\}$. Each exposure is a marginal measure, so the exposures of the ten vaults do not add up to the total duplicated amount. We relate exposure to where the vault lends, and the within-market duplication rate to who lends there,

$$D_{v,t} = \alpha_v + \alpha_t + \beta_1 Shared_{v,t} + \beta_2 \text{¿¿v},t + \beta_3 Markets_{v,t} + \varepsilon_{v,t} \text{¿}, \tag{5}$$

$$MDup_{m,t} = \alpha_m + \alpha_t + \beta_1 Vaults_{m,t} + \beta_2 Util_{m,t} + \beta_3 Cash_{m,t} + \beta_4 Top_{m,t} + \varepsilon_{m,t}, \tag{6}$$

where $Shared_{v,t}$ is the share of the vault's positions in markets that another sample vault uses, $Markets_{v,t}$ counts its markets, $MDup_{m,t}$ is the duplication rate within market *m* among the sample vaults that use it, $Vaults_{m,t}$ counts those vaults, $Cash_{m,t}$ is the log of the market's idle cash, $Top_{m,t}$ is the largest vault position divided by market supply, and α_v, α_m and α_t are vault, market and date effects. Equation (6) is estimated on the markets that at least two sample vaults use. Standard errors are clustered by vault and date in equation (5) and by market and date in equation (6). In the four-chain specifications, exposure is measured against all other vaults on the same chain, and both *Util* and *Top* are included, where *Top* is the share of the vault's largest position. Column (1) of Table 4 uses the September 2026 cross-section with chain effects, column (2) uses all seven dates with chain-by-date effects, and column (3) additionally includes vault effects. These regressions describe conditional associations among measures constructed

from shared accounting inputs. They do not identify behavioral or causal responses.

The withdrawal tests of Section 6 apply the individual and joint checks of Section 2 to requests for a fixed fraction of each vault's assets at every closing block. They then pay the same requests from the markets' recorded operations replayed forward from that block. Section 7 asks how a vault's capacity moves with the cash that others add or remove. For the day-to-day changes, we estimate

$$\Delta S_{v,t} = \alpha_v + \beta_1 Own_{v,t} + \beta_2 Peer_{v,t} + \beta_3 Outside_{v,t} + \beta_4 Borrow_{v,t} + \varepsilon_{v,t}, \tag{7}$$

where $\Delta S_{v,t}$ is the change in the vault's standalone capacity over the day. *Own* is the vault's own net supply into the sample markets. *Peer* is the net supply of the other nine sample vaults into the markets in which the vault holds a position at the opening block, *Outside* the net supply of all other suppliers into those markets, and *Borrow* net borrowing from them. Cash that others add to a market raises the vault's capacity when the vault is limited by the market's cash, and not when it is limited by its own position. The vault's own supply raises its position and the market's cash together, and borrowing removes cash. A second specification interacts *Peer* with *Shared*. Standard errors are clustered by date.

Large borrows are examined in event time. For each borrow *e* that takes at least a tenth of a market's idle cash, we observe the market at nine points in time *h*, from 24 hours before the borrow to one week after, and estimate

$$y_{e,h} = \alpha_e + \sum_h \beta_h 1(h) + \varepsilon_{e,h}, \tag{8}$$

where $y_{e,h}$ is the within-market duplication rate or the market's joint cash bound, α_e is an event effect, and 1(*h*) indicates the point in time. The state 24 hours before the borrow is the omitted category, so each β_h is the change relative to that state. Further specifications censor each event at the next large borrow in the same market, interact the indicators with the number of sample vaults that share the market, and repeat the estimation for placebo markets of similar size. Standard errors are clustered by event. The holder-level tests of Section 7 compare individual holders at the block before and the block of a deposit, using the expression $min\{d, (Q_j - M)^{+¿}¿$ of Section 2 and read-only executions of fixed requests against the archived states.

5. Scale and Persistence of the Joint Cash Gap

Table 2 reports the gap between what the ten vaults can withdraw separately and together. On the median day, the excess of summed standalone capacity over the joint cash bound is more than half of summed standalone capacity. Excluding Spark, the largest vault, or cbBTC/USDC, the market that holds most of the sample's idle cash, still leaves an excess of more than a third. In each market in which three or more sample vaults lend, the median within-market duplication rate is about one half. The dominant market differs in scale, not in the measured duplication mechanism. On the collection dates in September 2026, summed standalone

capacity matches the withdrawable-liquidity fields that the protocol's public interface reports, so our comparison uses the figures that depositors see.

Table 2. Duplicated capacity of the ten vaults

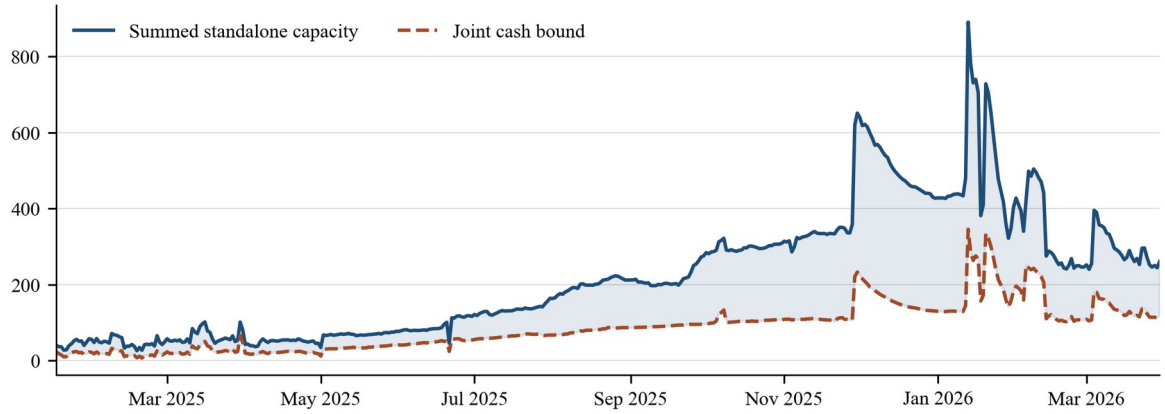
	Duplicated cash	Duplication (%)
All	115.9	57.86
Excluding Spark	34.21	44.85
Excluding cbBTC/USDC	1.295	35.18
Three vaults	29.52	42.17

Notes. Entries are medians over the 439 daily states of the ten native-USDC Morpho vaults on Base with the most assets on 16 January 2025. *Duplicated cash* is summed standalone capacity less the joint cash bound, in million USDC, and *Duplication* (%) is that difference in percent of summed standalone capacity. *All* uses the ten vaults, *Excluding Spark* drops the Spark vault, *Excluding cbBTC/USDC* drops that market, and *Three vaults* uses Spark, Moonwell Flagship, and Gauntlet Core.

Figure 1 plots the two quantities over the full sample. Both grow and fluctuate, but a large gap separates them on every date. The gap is not confined to stress windows.

Figure 1. Summed standalone capacity and the joint cash bound

Panel A. Summed standalone capacity and joint cash bound



Panel B. Duplicated share

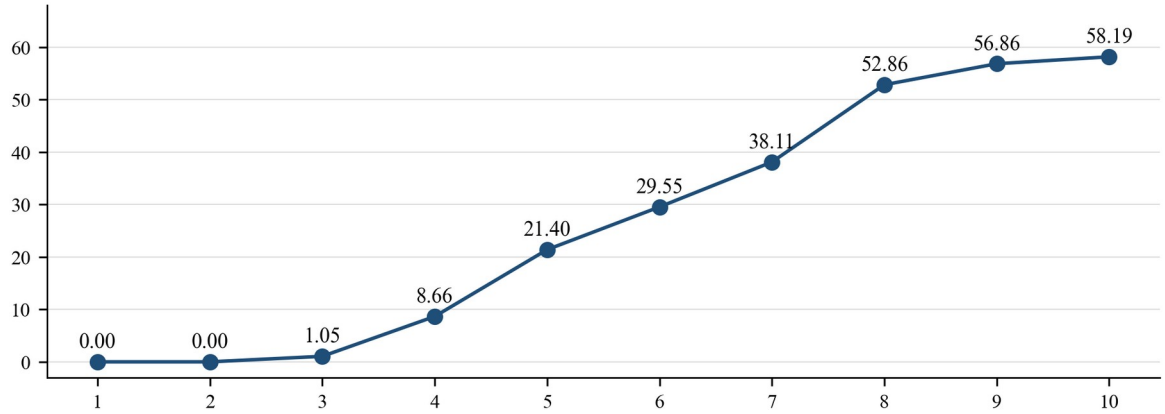


Notes. The horizontal axis is the daily closing date from 16 January 2025 to 30 March 2026 for the ten native-USDC Morpho vaults on Base with the most assets on the first day. In Panel A, the vertical axis is million USDC, and the shaded area is duplicated cash. In Panel B, the

vertical axis is duplicated cash in percent of summed standalone capacity.

Figure 2 shows how the duplication rate rises as vaults are added, using every subset of the ten vaults in the case window. In small subsets, combined positions rarely exceed a market's cash. The rate rises steeply as the fourth to eighth vaults are added and more slowly after that. Among subsets of the same size, the median rate is far higher for those that contain Spark. Subsets with the largest vault reach duplication with few members, whereas subsets without it need many vaults before their combined positions exceed a market's cash.

Figure 2. Duplication rate by number of vaults included



Notes. The horizontal axis is the number of vaults included, *k*, and the vertical axis is the duplication rate in percent of summed standalone capacity, for the ten native-USDC Morpho vaults on Base with the most assets on 16 January 2025. For each of the 319 dates to 30 November 2025, the median is taken across all subsets with exactly *k* members. The plotted value is the median of these daily medians, leaving out three subset-dates with zero summed standalone capacity.

Table B1 reports the same measurement for every USDC vault of the same contract version on Base and Ethereum on 29 September 2026. The gap is wider for all Base vaults than for the ten vaults, in line with Figure 2, and the duplication rate is about one half on Ethereum. Compared with assets, the difference is larger still. On Base, the vaults could separately withdraw about three fifths of their assets but together less than one fifth. On Ethereum, they could together withdraw less than one tenth. Appendix B repeats the calculation at month-ends for Morpho's USDC vaults on Arbitrum and Polygon, Euler Earn vaults on Ethereum and Yearn V3 vaults on Polygon. Duplication is large on Arbitrum, where dozens of vaults share the same markets, and small or absent on Euler Earn and Yearn, whose vaults rarely share a market. Duplication is therefore larger where more vaults share a pool, consistent with the market-level estimates in Table 3.

Table 3 reports the estimates of equations (4) to (6). In the daily series, the duplication rate is higher and the joint cash bound relative to assets is lower when a larger share of the vaults' markets is lent out (columns (1) and (2)). Higher utilization leaves less idle cash to set against the vaults' combined positions. Cash concentration and total assets are associated with a lower joint cash bound relative to assets but not with the duplication rate. The largest vault's net inflow is not associated with the joint cash bound. In column (3), a vault's exposure rises almost

one for one with the share of its positions in markets that other sample vaults also use. This estimate remains significant under a wild cluster bootstrap for the small number of vaults. Vault size and the number of markets show no association. In column (4), among markets that at least two sample vaults use, the within-market duplication rate rises with each additional vault and with the market's utilization, and the association with utilization is stronger in markets that more sample vaults share (Table B2). Idle cash and the size of the largest position show no association once market and date effects are included. The duplicated share of a vault's capacity is thus associated with where it lends and how many others lend there, not with its size.

Table 3. Determinants of duplication

	Dup	JA	D	MDup
	(1)	(2)	(3)	(4)
HHI	0.035	-0.064**		
	(1.17)	(-2.45)		
Size	-0.036	-0.039***		
	(-0.82)	(-3.14)		
Util	0.057***	-0.096***		
	(7.37)	(-8.15)		
Flow		0.041		
		(0.80)		
Shared			1.064***	
			(13.08)	
Size			0.001	
			(0.15)	
Markets			0.000	
			(0.08)	
Vaults				0.085***
				(8.47)
Util				0.039**
				(2.75)
Cash				-0.005
				(-0.65)
Top				-0.016
				(-0.19)
Year-month_FE	Yes	Yes	No	No
Weekday_FE	Yes	Yes	No	No
Vault_FE	No	No	Yes	No
Date_FE	No	No	Yes	Yes
Market_FE	No	No	No	Yes
R^2	0.787	0.741	0.663	0.191

Notes. An observation is a day in columns (1) and (2), a vault-day in column (3), and a market-day of a market with at least two sample vaults in column (4), for the ten Base vaults from 16 January 2025 to 30 March 2026, with 439, 4,387, and 3,167 observations after excluding zero standalone capacity or zero idle cash. The dependent variable is *Dup*, the daily duplication rate, in column (1), *JA*, the joint cash bound divided by assets, in column (2), *D*, each vault's exposure to the other nine vaults, in column (3), and *MDup*, the within-market duplication rate, in column (4), and columns (1) and (2) use previous-day explanatory variables, with *Flow* in

column (2) only. *Year-month_FE*, *Weekday_FE*, *Vault_FE*, *Date_FE*, and *Market_FE* denote year-month, weekday, vault, date, and market fixed effects. Values in parentheses are t-statistics calculated using Newey-West standard errors with a seven-day bandwidth in columns (1) and (2), standard errors clustered by vault and date in column (3), and standard errors clustered by market and date in column (4), and ** and *** denote significance at the 5% and 1% levels.

Table 4 extends the vault-level comparison to every native-USDC vault of Morpho's V1 contract on the four chains at seven dates from October 2025 to September 2026. Each vault's exposure is measured against all other vaults on its chain. Exposure rises with the share of a vault's positions in markets that other vaults also use and with the utilization of its markets. This holds in the September 2026 cross-section with chain effects, in the full sample with chain-by-date effects, and within the same vault over time. The associations with vault size, the number of markets and the concentration of positions are small and not consistent across specifications. In most vault-states, other vaults could take the whole of a vault's standalone capacity first, so the gap measured for the ten vaults is not a feature of a few large Base vaults. Leaving out any one chain, clustering standard errors by vault and by date, or dropping the vault-states with a nonzero loss ledger leaves the estimates on *Shared* and *Util* essentially unchanged (Table B4).

Table 4. Competing-cash exposure on four chains

	(1)	(2)	(3)
Shared	0.478***	0.541***	0.566***
	(9.88)	(13.32)	(10.29)
Markets	-0.009	0.005**	0.006
	(-0.76)	(1.97)	(1.48)
Size	-0.004*	-0.001	0.000
	(-1.78)	(-1.03)	(0.14)
Util	0.069***	0.068***	0.055***
	(16.14)	(13.97)	(9.61)
Top	-0.049	0.001	-0.013
	(-0.41)	(0.03)	(-0.25)
Chain_FE	Yes	No	No
Chain×Date_FE	No	Yes	Yes
Vault_FE	No	No	Yes
R^2	0.589	0.706	0.406

Notes. An observation is a vault-state of a native-USDC Morpho V1 vault with positive assets and positive standalone capacity on Ethereum, Base, Arbitrum, or Polygon, observed at the last block of each month from October 2025 to March 2026 and at a finalized block on 29 September 2026. The dependent variable is *D*, the share of a vault's standalone capacity that all other vaults on its chain could take first, *D* and *Shared* are measured against all other vaults on the chain, and all columns include *Util* and *Top*, the share of the vault's largest position. Column (1) uses the 338 vaults of the 29 September 2026 cross-section, columns (2) and (3) use all 2,141 vault-states, and *Chain_FE*, *Chain×Date_FE*, and *Vault_FE* denote chain, chain-by-date, and vault fixed effects. Values in parentheses are t-statistics calculated using heteroskedasticity-robust standard errors in column (1) and standard errors clustered by vault in columns (2) and (3), and *, **, and *** denote significance at the 10%, 5%, and 1% levels.

6. Individual Checks and the Joint Cash Bound

We now test withdrawal requests against a fixed date and a fixed set of vaults. Every vault receives a request equal to 5, 10 or 25 percent of its total assets. For fraction f, the individual check is $r_{vt} = fQ_{vt} \le W_{vt}$ for every vault, and the joint check fails when $\sum_{v} r_{vt} > J_t$. Failing the joint check shows that the requests cannot all be paid from the cash available at that date. Passing it is necessary for payment but not sufficient. Panel A of Table 5 shows how the joint check changes the assessment. Requests for 5 percent of assets pass every individual check on most dates and never exceed the joint cash bound. At 10 percent, only a handful of passing dates are excluded, by small amounts. At 25 percent, all ten vaults can meet their own request on fewer than one date in five. On about three quarters of those dates, the requests together exceed the joint cash bound, with a median shortfall of more than a quarter of the combined request. The joint cash bound thus adds information mainly for large requests, the ones that reach the cash several vaults count. Each vault's figure is accurate before other vaults withdraw, but it sets no cash aside, so the figures cannot all be honored at once when withdrawals are large relative to the pool. The joint cash bound is measured at one block, but cash keeps arriving. To see how long a shortfall lasts, we give every vault a request for a fixed fraction of its assets at each daily closing state and replay the recorded operations of the 33 markets forward. The ten vaults' own operations are removed, other participants' borrowing and withdrawals are capped at the cash that is left, and after each inflow the unpaid requests draw cash through each vault's withdrawal queue, largest vault first. Requests are divisible in the replay and are paid through successive partial withdrawals. Completion times are conditional on this allocation rule and are not observed waiting times of single withdrawal calls. Requests for 10 percent of assets are paid in full at the closing block on most dates and within 24 hours on almost all. Requests for a quarter of assets are paid at once on few dates. In this replay with fixed external flows, their median time to full payment is 128 hours, and on more than a third of dates they are still not fully paid after seven days (Panel B of Table 5). These dates form long spells in late summer and early winter of 2025, and payment takes no longer in the two stress windows (Figure B1). Payment at the closing block succeeds on fewer dates than Panel A counts as passing both checks, because each vault draws cash from its markets in the order of its withdrawal queue. At a quarter of assets, three of the four differences come from the order in which the vaults are paid. On the fourth date, 11 March 2025, the requests cannot all be paid at that block under any allocation.

Table 5. Individual and joint checks
Panel A. Checks at the closing block

	5%	10%	25%
Dates	439	439	439
Pass	413	353	81
Fail	0	6	62
Fail/Pass (%)	0.00	1.70	76.54
Both	413	347	19

Median deficit	-	0.848	55.33
Maximum deficit	-	3.623	120.3
Fail/Dates (%)	0.00	1.37	14.12
Deficit/Request (%)	-	1.25	28.71

Panel B. Time to full payment

	10%	25%
Immediate	313	15
Median hours	0.00	128.02
Paid 24h (%)	89.07	15.03
Unpaid 168h	7	170
Paid share (%)	99.98	93.25
New supply (%)	88.03	87.30
Repayments (%)	11.70	10.00

Notes. An observation is one of the 439 daily closing blocks, at which each of the ten native-USDC Morpho vaults on Base with the most assets on 16 January 2025 requests the fraction of its assets given by the column. In Panel A, *Dates* counts closing blocks, *Pass* counts dates on which every request is within the vault's own capacity, *Fail* counts passing dates on which the requests together exceed the joint cash bound, *Fail/Pass (%)* and *Fail/Dates (%)* give failing dates in percent of passing dates and of all dates, *Both* counts the remaining passing dates, *Median deficit* and *Maximum deficit* are the sum of requests less the joint cash bound over failing dates in million USDC, *Deficit/Request (%)* is the median across failing dates of each date's deficit in percent of its combined request, and a dash denotes an empty set. In Panel B, the requests are paid, largest vault first and through successive partial withdrawals, from the markets' recorded operations replayed forward from each closing block without the ten vaults' own operations, with other participants' recorded inflows held fixed and their borrowing and withdrawals capped at remaining cash, so completion times depend on this allocation rule and are not observed waiting times of single withdrawal calls. *Immediate* counts dates on which all requests are paid without later inflows, *Median hours* is the median time to full payment capped at 168 hours, *Paid 24h (%)* is the share of dates paid within 24 hours, *Unpaid 168h* counts dates with requests still unpaid after seven days, *Paid share (%)* is the median paid share on those dates, and *New supply (%)* and *Repayments (%)* are shares of recorded external inflows before completion, pooled over runs that reach full payment.

Two stress episodes fall inside the sample: the liquidation shock of 10 and 11 October 2025 and the losses at Stream Finance in early November 2025. We take 9 to 17 October and 3 to 21 November as the two windows and the 30 preceding days, excluding the other stress window, as comparison periods. For the ten Base vaults, the joint cash bound relative to assets is significantly lower in both windows, whereas the duplication rate does not differ. The large movements occur on the chains where the November losses fell. In hourly states of all native-USDC vaults, the joint cash bound falls to a few percent of assets on Ethereum and to almost nothing on Arbitrum, and it returns to its earlier median within about a day (Panel A of Table 6). Later in the November window, the duplication rate on both chains falls and the joint cash bound rises well above its earlier level (Figure 3). Vaults with no position in markets that accept the affected collateral show no consistent change in exposure. The transaction record also shows withdrawal calls that a vault did not execute. We scan every call to the 162 Base vaults and keep the withdrawal and redemption calls that failed with the vault's liquidity error. The

contract checks liquidity before it checks the owner's shares and the caller's allowance, so we keep only requests within the owner's claim at the preceding block, for claims of at least 100 USDC, and merge repeated calls by the same holder within an hour (Panel B of Table 6). Almost all events in the ten sample vaults occur in Universal USDC, a vault with assets below one million USDC, on dates in August and September 2025 when its idle cash is a tiny fraction of its assets. Holders whose call fails later complete a withdrawal of some amount from the same vault, after a median wait of about half a day during which other participants supply cash. The one event in a large vault is a request for the entire assets of Extrafi, followed within an hour by a completed withdrawal of part of that amount. Outside Base, such calls cluster in the November 2025 stress window on Ethereum and Arbitrum. During that window, 85 of 91 events on Ethereum and all 61 events on Arbitrum are followed by a completed withdrawal. The next withdrawal occurs within one hour in 63 and 45 events, respectively. These calls show that a vault's own cash constraint can bind in recorded transactions. They are calls that reverted with a liquidity error, not a general rate of failed withdrawals or of full recovery.

Table 6. Stress windows and reverted withdrawals

Panel A. Stress windows

	Base	Ethereum	Arbitrum
Bound, Oct (%)	10.91	12.18	14.86
Minimum, Oct (%)	10.70	10.58	1.88
Recovery, Oct	113	1	10
Bound, Nov (%)	11.79	11.91	11.35
Minimum, Nov (%)	10.43	3.13	0.64
Recovery, Nov	262	29	8
Errors, Oct	0 (11)	1 (0)	1 (24)
Errors, Nov	3 (0)	91 (0)	61 (0)

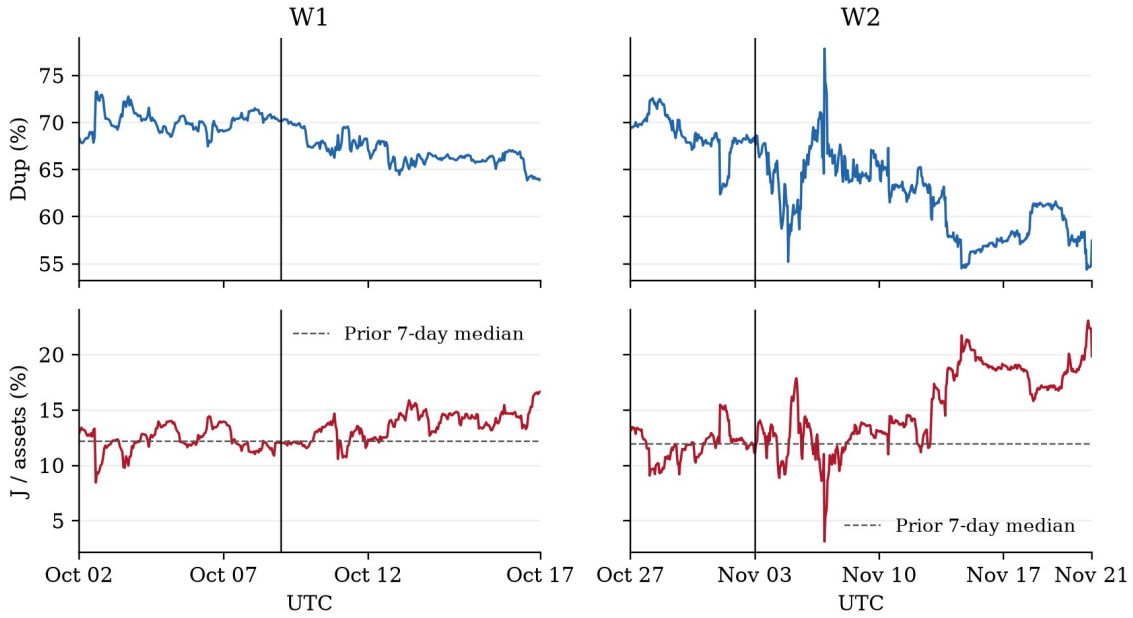
Panel B. Reverted withdrawal calls on Base

	Events	Dates	Holders	Request	Vault cash	Wait
Universal USDC	239	18	163	1,004	1,006	11.6
Extrafi XLend USDC	1	1	1	34,333,896	30,195,901	0.6
Other sample	0	0	0	-	-	-
Other Base	12	9	9	13,111	1,781	0.5
All vaults	252	28	173	1,005	1,006	10.7

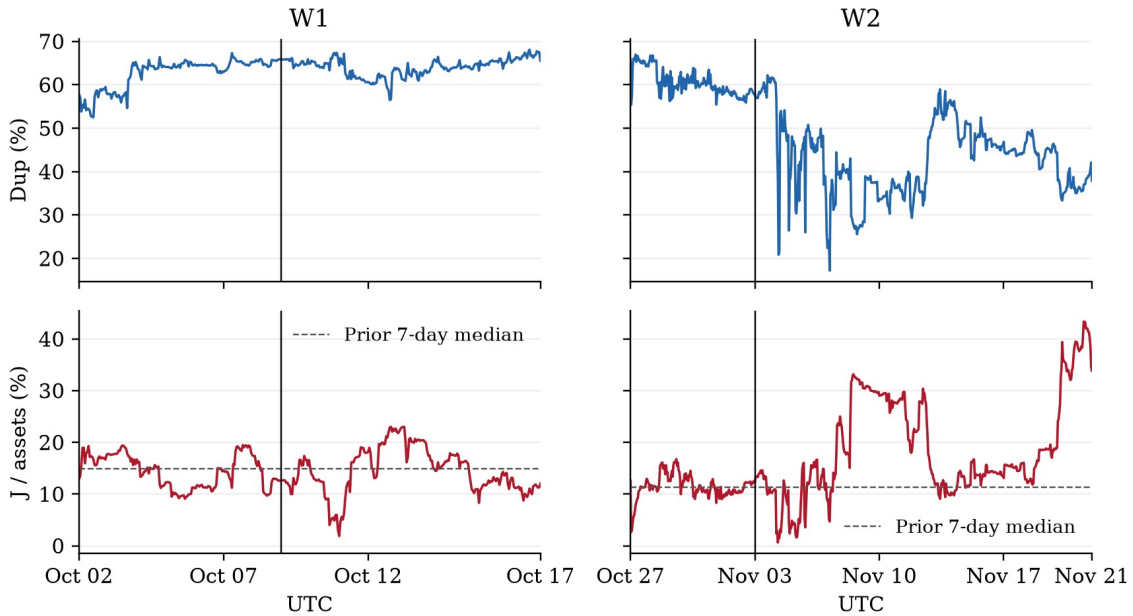
Notes. In Panel A, an observation is a chain in a stress window, the October window (*Oct*, 9 to 17 October 2025) or the November window (*Nov*, 3 to 21 November 2025), where Base covers the ten native-USDC Morpho vaults on Base with the most assets on 16 January 2025 and Ethereum and Arbitrum cover every native-USDC Morpho V1 vault with positive assets, read at the block closest to each UTC hour. *Bound* is the median joint cash bound in percent of vault assets over the seven days before the window, *Minimum* is its lowest hourly value in the window, *Recovery* is the number of hours from that minimum back to the earlier median, and *Errors* counts reverted withdrawal calls to all native-USDC V1 vaults of the chain under the filters of Panel B, with the count over the preceding 30 days, excluding the other window, in parentheses. In Panel B, an observation is an event, defined as a withdrawal or redemption call to the 162 native-USDC V1 vaults on Base from 16 January 2025 to 30 March 2026 that reverted with the vault's liquidity error and requested at least 100 USDC and no more than the

owner's claim at the preceding block, with calls by the same owner on the same vault within one UTC hour merged into one event. *Events*, *Dates*, and *Holders* count events, dates, and distinct holders, *Other sample* and *Other Base* are the remaining sample vaults and the other Base vaults, *Request* and *Vault cash* are the median requested amount and the median withdrawable cash of the vault at the preceding block in USDC, and *Wait* is the median time in hours to the owner's next completed withdrawal of any amount from the same vault, excluding one unresolved event, so it does not establish full recovery of the original request.

Figure 3. Duplication and the joint cash bound in stress windows
Panel A. Ethereum



Panel B. Arbitrum



Notes. Panel A covers every native-USDC Morpho V1 vault with positive assets on Ethereum, and Panel B covers the same population on Arbitrum, read at the block closest to each UTC hour. The left column covers 2 to 17 October 2025, and the right column covers 27 October to

21 November 2025. The vertical line marks the start of the October window ($W1$) and the November window ($W2$). The upper rows show the duplication rate in percent of summed standalone capacity, and the lower rows show the joint cash bound in percent of the vaults' assets.

7. Cash Replenishment and Withdrawal Capacity

We first look across the vaults that share one market. At Base block 36,399,032 on 4 October 2025, Spark makes a large supply to the cbBTC/USDC market that doubles its idle cash, with no other event in that market during the block. Table 7 compares the nine sample vaults with a position in the market just before and after the block. Two vaults gain capacity although neither their positions nor their outstanding shares change. Moonwell gains part of the deposit and reaches the limit set by its own position. Steakhouse gains the full deposit and is still limited by the market's cash. The other six are already limited by their own positions. These gains are several claims on the same new cash. Within the market, summed standalone capacity rises by more than twice the deposit, while the joint cash bound rises only by the amount deposited. A deposit thus eases the limit of the vaults that are short of market cash and enlarges the duplicated amount at the same time.

Table 7. Vaults at the replenishment block

	Unchanged	Position before	Position after	Capacity before	Capacity after	Capacity change
Moonwell Flagship	Yes	55.94	55.94	48.36	55.94	7.580
Extrafi XLend	Yes	6.776	6.776	6.998	6.998	0.000
Re7	Yes	0.414	0.414	0.813	0.813	0.000
Gauntlet Core	Yes	3.857	3.857	4.029	4.029	0.000
Spark	No	579.0	627.8	48.36	97.21	48.85
Gauntlet Prime	Yes	22.39	22.39	22.64	22.64	0.000
Steakhouse	Yes	190.3	190.3	48.76	97.62	48.85
Ionic Ecosystem	Yes	0.030	0.030	0.030	0.030	0.000
Universal	Yes	0.986	0.986	1.169	1.169	0.000

Notes. An observation is one of the nine sample vaults with a position in the cbBTC/USDC market at Base blocks 36,399,031 and 36,399,032 on 4 October 2025, and amounts are in million USDC. *Unchanged* states whether the vault's supply shares in the market are identical at the two blocks. *Position before* and *Position after* are its accrued positions in this market, *Capacity before* and *Capacity after* are its standalone capacity over its full withdrawal queue, and *Capacity change* is their difference. Entries shown as 0.000 lie between 0.0003 and 0.234 USDC.

Table 8 asks whether the full sample shows the same dependence on other participants' cash. Capacity rises with inflows from every source and falls with borrowing. In column (1), cash from other sample vaults and cash from outside suppliers move capacity by amounts that are not statistically different. A vault's withdrawal capacity thus may be refilled by whoever adds cash to its markets and drained by whoever borrows from them. Column (2) shows that the association with inflows from other sample vaults is stronger when a larger share of the vault's positions is in markets those vaults use. For a vault with no shared positions, the peer coefficient

is not distinguishable from zero.

Table 8. Others' cash and withdrawal capacity

	(1)	(2)
Own	0.637***	0.645***
	(4.67)	(4.75)
Peer	0.185***	-0.029
	(7.62)	(-0.38)
Outside	0.228***	0.226***
	(33.94)	(41.37)
Borrow	-0.394***	-0.387***
	(-3.52)	(-3.47)
Peer×Shared		0.241***
		(2.63)
Shared		-0.091
		(-0.15)
p(Peer=Outside)	0.116	0.632
Vault_FE	Yes	Yes
R^2	0.296	0.304

Notes. An observation is one of the 4,380 vault-days of the ten Base vaults, and the dependent variable is ΔS, the daily change in the vault's standalone capacity in million USDC. *Own* is the vault's own net supply, *Peer* the net supply of the other nine sample vaults, *Outside* the net supply of all other suppliers, and *Borrow* net borrowing, each in million USDC in the markets in which the vault holds a position at the opening block, *Shared* is the share of the vault's positions in markets that another sample vault uses, and *Peer×Shared* interacts *Peer* with *Shared*. *p(Peer=Outside)* is the p-value for equal coefficients on *Peer* and *Outside* in column (1), and for *Peer+Peer×Shared* at *Shared*=1 equal to *Outside* in column (2), and *Vault_FE* denotes vault fixed effects. Values in parentheses are t-statistics calculated using standard errors clustered by date, and *** denotes significance at the 1% level.

Borrowing works in the opposite direction. Table 9 estimates equation (8) for the borrows that take at least a tenth of a market's idle cash in one transaction, keeping the first such borrow in any 24-hour window. Compared with 24 hours before, the market's duplication rate is higher at the block of the borrow, and its joint cash bound falls sharply (column (1)). Large borrows come in clusters. Another borrow above the threshold follows in the same market within 24 hours for almost half of the events and within a week for four fifths, so the later coefficients in column (1) also reflect the borrows that follow. Column (2) truncates each event at the next such borrow. The estimates at the block of the borrow are unchanged. At the 10 percent level, the duplication rate is still higher one hour later, and the joint cash bound stays lower for six hours. At 24 and 72 hours, neither estimate is distinguishable from zero relative to the state 24 hours before the borrow, and at 168 hours, the duplication rate is lower at the 10 percent level in the events that remain. These later estimates condition on the absence of a subsequent large borrow, so they describe the retained event paths rather than a lasting effect of the first borrow. For the events with no other borrow above the threshold within 168 hours before or after, the fall in the joint cash bound at the block is significant at the 10 percent level, the rise in the duplication rate is not, and no later coefficient is distinguishable from zero. Both responses are

larger where more sample vaults share the market (column (3)). Under a wild cluster bootstrap by market with Rademacher weights and 9,999 draws, both changes at the block remain significant at the 5 percent level. Placebo markets of similar size show no significant change at the event block, although their duplication rate declines at some later horizons (column (4)). A large borrow is thus associated with changes in the two measures at the block and for some hours afterwards, and the longer-lasting differences in column (1) come from the borrows that follow.

Table 9. Large borrows

Panel A. Duplication rate

	(1)	(2)	(3)	(4)
-1 hour	-0.002	-0.002	0.003	-0.004
	(-0.54)	(-0.53)	(1.35)	(-1.52)
Block before	-0.001	-0.001	0.005*	-0.004
	(-0.23)	(-0.23)	(1.80)	(-1.40)
Event block	0.016***	0.016***	0.014***	-0.004
	(3.62)	(3.54)	(4.73)	(-1.42)
+1 hour	0.014***	0.009*	0.008***	-0.004
	(3.00)	(1.96)	(2.73)	(-1.37)
+6 hours	0.012**	0.007	0.007**	-0.005
	(2.43)	(1.43)	(2.20)	(-1.49)
+24 hours	0.013**	0.004	0.007**	-0.010***
	(2.41)	(0.73)	(2.04)	(-2.60)
+72 hours	0.012*	-0.009	0.007	-0.007
	(1.85)	(-0.95)	(1.58)	(-1.49)
+168 hours	0.007	-0.024*	0.003	-0.009*
	(1.19)	(-1.77)	(0.69)	(-1.67)
Event_FE	Yes	Yes	Yes	Yes
R^2	0.008	0.018	0.014	0.004

Panel B. Joint cash bound

	(1)	(2)	(3)	(4)
-1 hour	-0.045*	-0.045	-0.068**	0.128
	(-1.65)	(-1.62)	(-2.07)	(1.04)
Block before	-0.070**	-0.070**	-0.085**	0.149
	(-2.34)	(-2.29)	(-2.11)	(1.20)
Event block	-0.225***	-0.225***	-0.403***	0.148
	(-3.43)	(-3.35)	(-3.19)	(1.19)

+1 hour	-0.090**	-0.093*	-0.112	0.147
	(-2.18)	(-1.95)	(-1.58)	(1.21)
+6 hours	-0.099**	-0.102*	-0.143*	0.173
	(-2.27)	(-1.94)	(-1.93)	(0.84)
+24 hours	-0.066	-0.060	-0.109	-0.019
	(-0.92)	(-0.52)	(-0.75)	(-1.19)
+72 hours	-0.042	-0.050	-0.088	-0.016
	(-0.63)	(-0.37)	(-0.69)	(-0.90)
+168 hours	0.010	-0.045	-0.020	-0.015
	(0.13)	(-0.18)	(-0.13)	(-0.80)
Event_FE	Yes	Yes	Yes	Yes
R^2	0.005	0.004	0.036	0.002

Notes. An observation is an event at one point in event time, where events are the 834 borrows from 16 January 2025 to 30 March 2026 in the 33 markets of the ten Base vaults that exceed 10 percent of the market's idle cash at the preceding block, keeping the first such borrow in any 24-hour window, and the rows give the time relative to the borrow, with *Block before* and *Event block* the blocks before and of the borrow. The dependent variable is *MDup*, a fraction, in Panel A and *J*, in million USDC, in Panel B, each coefficient is the difference from the state 24 hours before the event, and because *MDup* is undefined where summed standalone capacity is zero, the samples have 5,859, 4,335, 5,859, and 3,924 observations in Panel A and 7,506, 5,514, 7,506, and 4,617 in Panel B. Columns (1), (3), and (4) use events with all nine states observed, column (2) drops observations at or after the next borrow above the threshold in the same market, column (3) interacts the indicators with the demeaned number of sample vaults sharing the market at the block before, column (4) uses the market of the same supply decile with the nearest supply and no borrow on that date, and *Event_FE* denotes event fixed effects. Values in parentheses are t-statistics calculated using standard errors clustered by event, and *, **, and *** denote significance at the 10%, 5%, and 1% levels.

The same block also changes what a second holder inside the Spark vault can withdraw. Two publicly identified holders dominate this vault. The first is the Spark treasury, identified by Spark's own registry, whose wallet records link funds that arrived from other chains to 94 vault deposits from April to November 2025 (Appendix A).[3] The second is Ethena, identified by its custodian's attestation.[4] We call the ratio of a holder's withdrawal capacity to its claim its coverage. Ethena's coverage is below 100 percent in more than a third of its eligible opening states and falls to less than half at its lowest. In the 4 October block, Ethena's shares do not change, yet its capacity rises and its coverage returns to 100 percent. Panel A of Table 10 reports three treasury deposits funded by cross-chain arrivals that exceed the market's cash while Ethena holds shares. For each, we run a withdrawal request equal to Ethena's full claim

[3] Spark address registry, commit 907aa003, 26 November 2025, https://github.com/sparkdotfi/spark-address-registry/tree/907aa003055069e9d17397873ccdae433f761b72/src; Circle, Cross-Chain Transfer Protocol mainnet launch announcement and developer documentation, 2023, https://investor.circle.com/news/news-details/2023/Circle-Delivers-USDC-Interoperability-Across-Ecosystems-with-Mainnet-Launch-of-Cross-Chain-Transfer-Protocol/default.aspx.

[4] Ethena Labs, Custodian attestations of assets backing USDe, August 2025, published 29 August 2025, https://paragraph.com/@ethena-labs/custodian-attestations-of-assets-backing-usde-august-2.

before and after the deposit. The request fails before the 4 October block and succeeds after it. The smaller deposit twenty minutes earlier raises capacity but not to the full claim. At the third deposit, Ethena is already fully covered and nothing changes, as the expression in Section 2 implies. Figure 4 shows the sequence: Ethena completes three large withdrawals within about 17 minutes, two treasury deposits follow within half an hour, and after the second, the test request for Ethena's full claim succeeds.

Table 10. Holder capacity around treasury deposits

Panel A. Ethena's full-claim request

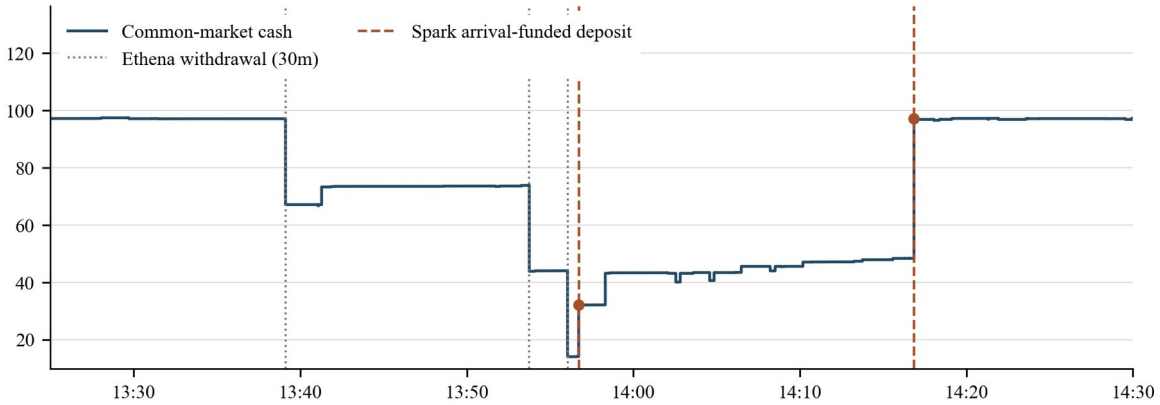
	Deposit	Capacity before	Capacity after	Fixed request
10-04 / 36398428	18.00	14.09	32.09	Fail / Fail
10-04 / 36399032	48.85	48.36	61.07	Fail / Pass
10-05 / 36426295	46.29	1.115	1.115	Pass / Pass

Panel B. Other holders around matched deposits

	All eligible	Constrained	Covered
Obs	495	17	478
Transactions	93	17	93
Holders	38	1	38
Capacity gains	17	17	0
Full claim	1	1	0
Clean gains	12	12	0
Median gain	0.011	1,339,230.193	0.009

Notes. In Panel A, an observation is one of the three treasury deposits in 2025, listed by date and Base block, that exceed the cbBTC/USDC market's cash while Ethena holds shares, with amounts in million USDC, where *Deposit* is the deposit amount, *Capacity before* and *Capacity after* are Ethena's maximum withdrawal at the block before and the block of the deposit with its shares unchanged, and *Fixed request* reports the read-only execution, before and after the deposit, of a withdrawal of Ethena's pre-deposit full claim, with *Fail* the contract's liquidity error and *Pass* a successful execution. In Panel B, an observation is a holder-transaction pair for Ethena and the five largest other holders by the previous day's claim in each of the 93 transactions that contain the 94 matched treasury deposits, with a positive opening claim and unchanged shares. *Constrained* holders open with a claim that exceeds capacity by more than one USDC, *Covered* holders do not, *All eligible* pools the two, *Obs* counts holder-transaction pairs, and *Transactions* and *Holders* count the transactions and holders involved. *Capacity gains* counts observations whose capacity rises by more than one USDC, *Full claim* counts constrained observations whose capacity after the deposit reaches the opening full claim, *Clean gains* counts gains in blocks with no other operation in the market, and *Median gain* is in USDC.

Figure 4. Withdrawals and deposits on 4 October 2025



Notes. The horizontal axis is UTC time on 4 October 2025, and the vertical axis is idle cash in the cbBTC/USDC market of Morpho on Base in million USDC, stepped at every supply, withdrawal, borrow, and repayment. Circles mark the market's cash after each deposit.

The 4 October episode is not isolated. For each transaction that contains a matched deposit, we observe Ethena and the five largest other holders at the block before and the block of the deposit (Panel B of Table 10). In a small number of observations, the holder starts more than one USDC short of full coverage. Every one of these gains capacity, while none of the fully covered observations gains more than one USDC, as the expression in Section 2 implies. Where the block contains no other operation in the common market, the change in capacity equals the smaller of the deposit and the holder's opening shortfall to the micro-USDC in all but one case. The contracts thus execute the holder-level accounting of Section 2. All constrained observations belong to Ethena, whose wallet contains no funds bridged from another chain, so the holder whose capacity the treasury's deposits restore is funded in a different way. The extra capacity lasts until other participants use the cash. One and six hours after the deposit block, every constrained position keeps its gain, and after 24 hours, 11 of the 12 positions whose shares are unchanged still hold a gain, but none covers its full claim. The treasury's own requests show the limit from the other side. Its median coverage is low, and a request for a quarter of its claim exceeds capacity on most days. Its completed withdrawals are small and never exceed a quarter of the prior claim in one transaction. When the vault's positions and the treasury's claim are scaled down to a quarter of their size with market cash held fixed, the same request would fail on a single day. What makes the treasury's coverage low is its own size relative to the cash that others share (Appendix C).

8. Conclusion

Lending vaults that share markets can each report a correct withdrawal capacity, while the sum of these capacities exceeds the cash they could withdraw together. The results identify a non-additivity problem in product-level liquidity measurement. For the ten Base vaults, the excess is more than half of summed standalone capacity on the median day, and a similar gap appears on the other chains where several vaults share markets. The gap matters for large requests. At a quarter of assets, requests that every vault could meet on its own exceed the joint cash bound on most of the dates on which all individual checks pass. Satisfying the aggregate bound does not by itself ensure that a particular set of requests can be paid, because each vault

draws cash in the order of its withdrawal queue. A deposit into a shared market raises the capacity of vaults and holders whose own positions do not change, and a vault's capacity moves with the cash that others supply to or borrow from its markets. Standalone figures therefore show how much cash each vault could reach, but not how much of the cash behind immediate withdrawals the vaults share. A useful disclosure is specific to the state and to the portfolio, identifying the selected vaults, their shared markets and the cash constraints relevant to joint withdrawals. Withdrawal calls that reverted with the liquidity error are concentrated in one small vault and show that a vault's own cash constraint can bind in recorded transactions, whereas the joint cash bound asks whether large simultaneous requests that pass individual checks could all be paid from the cash available at a common state. The replay describes payment under fixed external flows and repeated partial withdrawals, and it does not estimate run probabilities, equilibrium withdrawal delays or the welfare effect of changing disclosure. The evidence comes mainly from one protocol, and whether large withdrawals from one vault are followed by withdrawals from other vaults that share its markets is a question for future work.

Appendix A. Attributing the treasury's deposits to bridged funds

CCTP moves native USDC between chains by burning it on the source chain and minting it on the destination chain. How much of each vault deposit was paid with bridged funds is bounded from the receiving wallet's ordered record of transfers. Let D_i be the amount of matched deposit i and $[L_i, U_i]$ the range of that amount that could have been paid with bridged funds. The range is built from the wallet's opening cash, each matched mint, its other incoming transfers and its outgoing transfers in transaction order. A deposit counts as matched when bridged funds arrive and leave for the vault within the same transaction, and funds that leave the wallet and later return count as other cash, so no outflow is attributed twice.

$$0 \le q_k \le x_k, 0 \le F_k - \sum_{j \le k} q_j \le B_k, \tag{A1}$$

where $F_k - \sum_{j \le k} q_j$ is the stock of matched funds still in the wallet after outflow k. The first inequality says that no outflow carries more matched funds than its own amount. The second says that matched funds cannot be spent before they arrive and cannot exceed the cash actually in the wallet. Other incoming transfers enter through B_k as other cash. For a group G of deposits, the lower and upper bounds are the minimum and maximum of $\sum_{k \in G} q_k$ over all feasible assignments, with every other outflow still constrained. Each transaction has its own feasible set, so the bounds add up across transactions. Within one transaction, the bounds for different deposits need not be attainable at the same time. The reported bounds use assignments in whole micro-USDC, and checks with a continuous linear program agree to within one micro-USDC.

Table A1 reports the result. Almost all of the value of the matched deposits could have been paid with bridged funds, and the deposits that are identified exactly carry the bulk of that value, so the conclusion does not depend on how the uncertain cases are resolved.

Table A1. Bridged funding of matched deposits

	Obs	Deposits	Value (%)	Lower	Upper
All deposits	94	490.8	100.00	471.8	489.9
Point identified	53	463.5	94.44	462.6	462.6
Positive-width range	41	27.29	5.56	9.184	27.29
Zero lower bound	35	14.97	3.05	0.000	14.97

Notes. An observation is one of the 94 matched deposits of the Spark treasury from April to November 2025, and amounts are in million USDC. *Obs* counts deposits, *Deposits* is their value, and *Value (%)* is its share of the total. *Lower* and *Upper* are the smallest and largest amounts that could have been paid with bridged funds under the constraints of equation (A1), optimized jointly for each group. *Point identified* deposits have equal bounds, *Positive-width range* deposits do not, and *Zero lower bound* is the subset of the latter whose lower bound is zero.

Appendix B. Additional results

Table B1 reports standalone capacity and the joint cash bound for every native-USDC Morpho V1 vault with positive assets on Base and Ethereum on 29 September 2026, discussed in Section 5.

Table B1. All vaults on Base and Ethereum

	Base	Ethereum
Vaults	162	146
Markets	79	121
Assets	1,006.331	811.2
Standalone capacity	606.2	158.5
Joint cash bound	181.6	77.17
Duplicated cash	424.7	81.31
Duplication (%)	70.05	51.31
Standalone/Assets (%)	60.24	19.54
Joint/Assets (%)	18.04	9.51
Block	51953448	26084069
Time (UTC)	2026-09-29 15:37:23	2026-09-29 15:37:59

Notes. An observation is a chain at the stated *Block* and *Time (UTC)*, covering every native-USDC Morpho V1 vault with positive assets on Base or Ethereum. *Vaults* counts the vaults, *Markets* counts the markets in which at least one of them holds a position, and *Assets* is the sum of their total assets. *Standalone capacity* and *Joint cash bound* follow equations (1) and (2), and *Duplicated cash* is their difference, all in million USDC. *Duplication (%)* is that difference in percent of standalone capacity, and *Standalone/Assets (%)* and *Joint/Assets (%)* divide the two quantities by assets.

Table B2 asks whether utilization matters more where more vaults share a market. In the market-day sample of the ten Base vaults, the association between utilization and the within-market duplication rate becomes stronger as the number of sample vaults in the market rises, and when the markets are split at the median of three vaults, the utilization coefficient is more than twice as large in the markets with more vaults. The coefficient on idle cash is not stable across the columns, so the result is an interaction between a market's utilization and the number

of vaults that draw on it, and not a direct effect of low cash balances.

Table B2. Utilization and the number of vaults

	(1)	(2)	(3)
Vaults	0.002	0.121***	0.069**
	(0.05)	(4.87)	(2.72)
Util	0.012	0.031*	0.077***
	(0.73)	(2.08)	(8.93)
Util×Vaults	0.010**		
	(2.48)		
Cash	-0.004	-0.008	0.030**
	(-0.60)	(-0.98)	(3.08)
Top	-0.022	0.013	-0.236
	(-0.26)	(0.13)	(-1.46)
Market_FE	Yes	Yes	Yes
Date_FE	Yes	Yes	Yes
R^2	0.197	0.100	0.278

Notes. An observation is a market-day of a market that at least two of the ten Base vaults use, the dependent variable is *MDup*, and *Util*×*Vaults* interacts *Util* with *Vaults*. Column (1) uses all 3,167 market-days, column (2) the 1,725 market-days with at most three sample vaults, and column (3) the 1,442 with more than three, and *Market_FE* and *Date_FE* denote market and date fixed effects. Values in parentheses are t-statistics calculated using standard errors clustered by market and date, and *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table B3 applies the definitions of Section 2 to every USDC vault of Morpho's V1 contract on Arbitrum and on Polygon, every Euler Earn vault on Ethereum and every Yearn V3 USDC vault on Polygon, each read at one block per date. For each group, we construct the closest contract-level analogue of standalone capacity and the joint cash bound from the vaults' positions in each underlying pool and the pool's idle cash. Because the rules by which a strategy recovers funds differ across protocols, the Euler and Yearn measures are narrower than the Morpho measure and serve as supporting evidence.

Table B3. Other protocols and chains

	Morpho V1 Arbitrum	Morpho V1 Polygon	Euler Earn Ethereum	Yearn V3 Polygon
Vaults	42	15	11	3
Markets	23	11	14	1
Assets	13.7	0.5	3.5	0.1
Duplication rate (%)				
31 October 2025	59.3	0.0	12.9	0.0
30 November 2025	56.3	33.8	2.3	0.0
31 December 2025	54.3	30.7	7.5	0.0
31 January 2026	61.8	26.8	10.0	0.0
28 February 2026	64.3	8.3	0.0	0.0
31 March 2026	50.0	0.0	0.0	0.0
29 September 2026	27.1	0.0	0.0	0.0

Notes. An observation is a protocol-chain population of every vault of the stated protocol and chain with positive assets, and entries under *Duplication rate (%)* are duplication rates in percent of summed standalone capacity at the stated dates. *Vaults*, *Markets*, and *Assets* describe the population on 29 September 2026, with assets in million USDC. Morpho V1 positions and market cash are read from the vault queues and market balances, and Euler Earn positions are the expected supply to each strategy vault, with that vault's idle balance as cash. Yearn V3 positions are each strategy's current debt, and cash is the idle USDC held by the strategy or by the common liquidity contract through which it lends, plus the vault's own idle balance counted once.

Table B4 re-estimates the specification with vault effects in Table 4, leaving out each chain in turn and, in the last column, clustering standard errors by vault and by observation date. Exposure is associated with shared-market positions and with utilization on every subset of chains and under both clustering choices. The loss ledger of the V1.1 contract is nonzero in a small share of the vault-states, and dropping those states leaves the estimates on *Shared* and *Util* essentially unchanged.

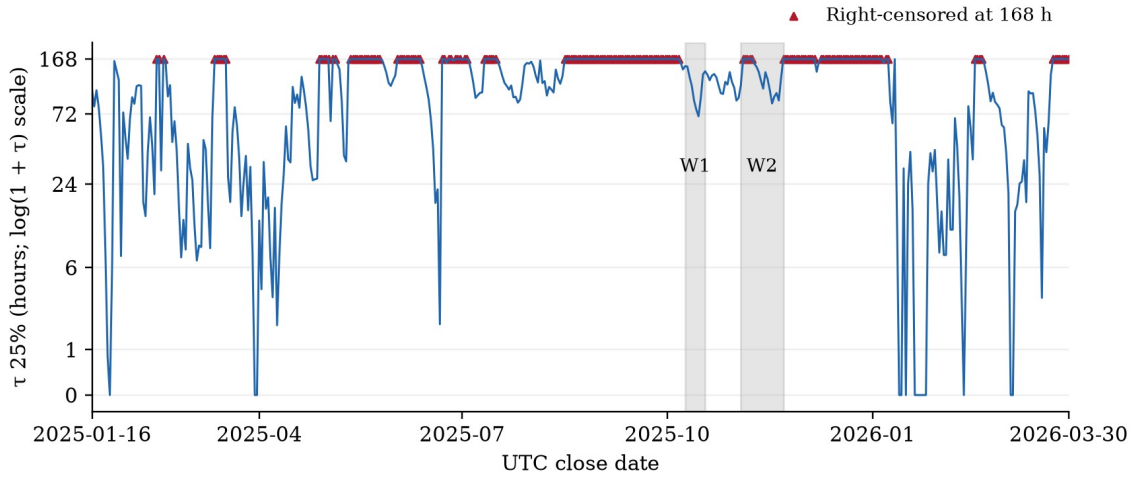
Table B4. Exposure on subsets of chains

	(1)	(2)	(3)	(4)	(5)
Shared	0.526***	0.588***	0.573***	0.563***	0.566***
	(6.24)	(9.29)	(9.51)	(10.12)	(10.71)
Markets	0.009	0.002	0.008*	0.007	0.006
	(1.09)	(0.54)	(1.68)	(1.52)	(1.35)
Size	0.000	0.002	0.000	0.000	0.000
	(-0.10)	(0.48)	(0.04)	(0.02)	(0.18)
Util	0.055***	0.059***	0.055***	0.053***	0.055***
	(8.13)	(7.74)	(8.67)	(8.84)	(5.96)
Top	0.011	-0.028	-0.011	-0.014	-0.013
	(0.15)	(-0.51)	(-0.20)	(-0.26)	(-0.35)
Vault_FE	Yes	Yes	Yes	Yes	Yes
Chain×Date_FE	Yes	Yes	Yes	Yes	Yes
R^2	0.367	0.447	0.406	0.400	0.406

Notes. An observation is a vault-state of the four chains, and the dependent variable is *D*, measured against all other vaults on the vault's chain. Columns (1) to (4) drop the vaults on Ethereum, Base, Arbitrum, and Polygon in turn, with 1,287, 1,184, 1,905, and 2,047 vault-states, column (5) uses all 2,141 vault-states, and *Vault_FE* and *Chain×Date_FE* denote vault and chain-by-date fixed effects. Values in parentheses are t-statistics calculated using standard errors clustered by vault in columns (1) to (4) and by vault and by the seven observation dates in column (5), and * and *** denote significance at the 10% and 1% levels.

Figure B1 reports the refill times discussed in Section 6. Every reconstructed vault state reconciles to its contract total and every market's cash to its recorded operations to the micro-USDC.

Figure B1. Time to full payment at a quarter of assets



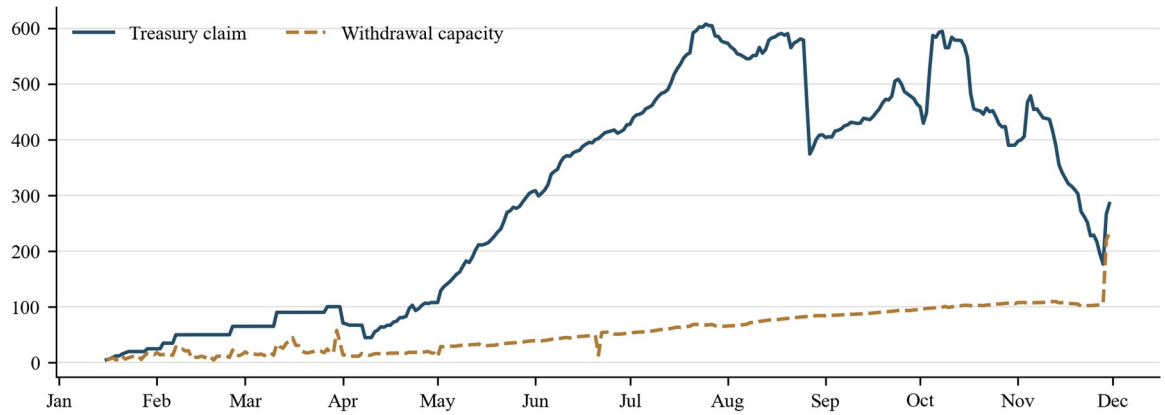
Notes. The horizontal axis is the daily closing date from 16 January 2025 to 30 March 2026 for the ten native-USDC Morpho vaults on Base with the most assets on 16 January 2025, and each run is followed for at most 168 hours. The vertical axis is the time in hours until requests for a quarter of every vault's assets are paid in full from recorded inflows, on a log(1+hours) scale, with requests paid at the closing block at zero. Shaded areas mark the October (*W*1) and November (*W*2) windows.

Appendix C. The treasury's withdrawals and the cash coverage of other holders

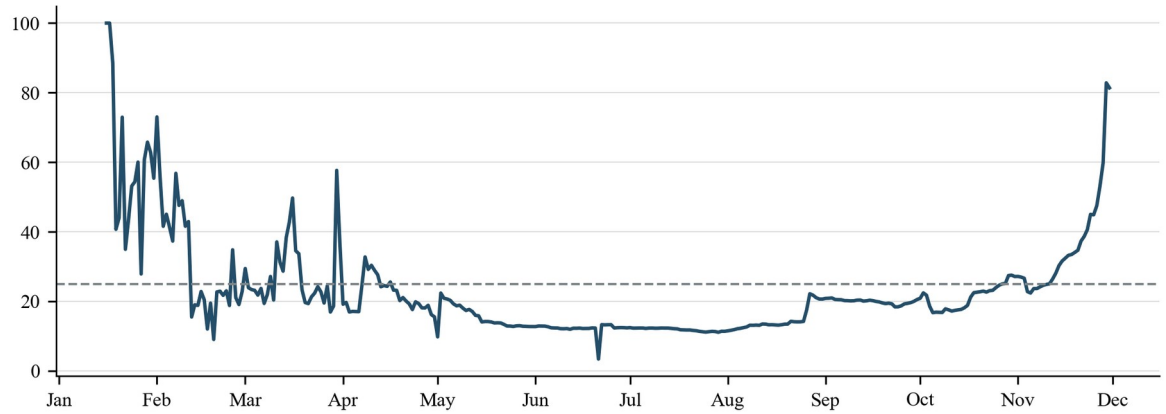
Withdrawal limits in the Spark vault fall on large positions. The treasury holds almost all of the vault's shares, and its withdrawable cash is a small fraction of its claim on the median day (Figure C1). Panel A of Table C1 compares requests for fixed fractions of the treasury's claim with what the treasury actually withdraws: requests for 5 and 10 percent almost never exceed capacity, a request for a quarter exceeds capacity on most days, and the median completed withdrawal is a tiny fraction of the previous day's claim, with no single transaction above a quarter of it. The same account therefore makes many small withdrawals and faces a binding limit on one large immediate request.

Figure C1. Treasury claim and withdrawal capacity

Panel A. Claim and available withdrawal



Panel B. Coverage



Notes. The horizontal axis is the daily closing date from 16 January to 30 November 2025 for the Spark treasury's position in the Spark vault on Base. In Panel A, the vertical axis is million USDC. In Panel B, the vertical axis is coverage, withdrawal capacity divided by the claim, in percent, and the horizontal line marks 25 percent.

Table C1. Treasury requests and other holders

Panel A. Fixed request fractions

	Capacity exceeded	Transactions above	Daily totals above
5%	1/319	17/12,321	112/319
10%	3/319	2/12,321	48/319
25%	247/319	0/12,321	8/319
50%	300/319	0/12,321	0/319
100%	317/319	0/12,321	0/319

Panel B. Other holders by claim size

	Owner-days	Addresses	Withdraw (%)	Deposit (%)	Cash<Q
1 to 99	289,662	7,832	1.76	2.25	0
100 to 9,999	180,437	4,932	2.94	1.68	0
10,000 to 999,999	42,856	1,525	5.77	3.14	0
≥1,000,000	994	42	14.59	9.96	39

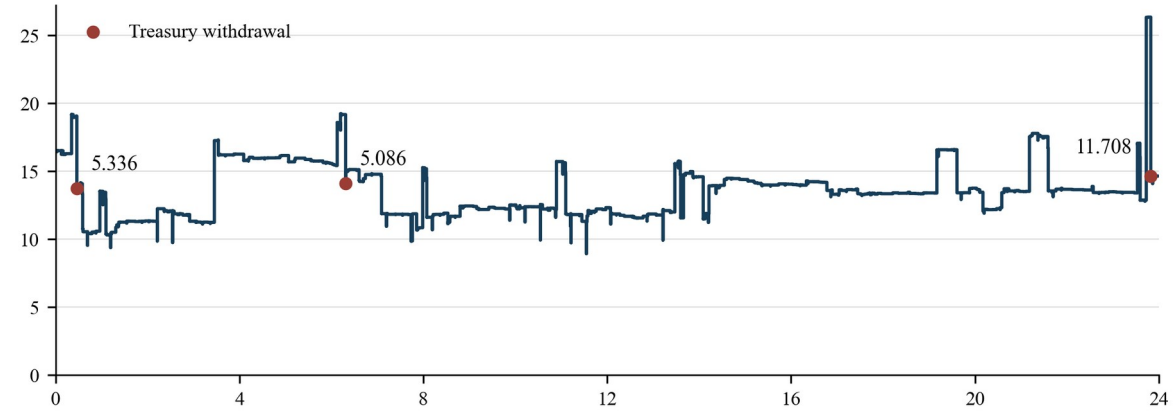
Notes. Both panels use the Spark vault on Base from 16 January to 30 November 2025. In Panel A, an observation is a fraction of the treasury's closing claim, *Capacity exceeded* counts the daily states on which the fraction exceeds the treasury's withdrawal capacity, with 319 archived read-only executions and 247 liquidity reverts in the 25% row, and *Transactions above* and *Daily totals above* count completed withdrawal transactions and daily withdrawal totals above the fraction of the prior closing claim, among 12,321 transactions and 319 days. In Panel B, an observation is a group of the other holders' daily opening claims Q in USDC, *Owner-days* counts holder-days with a claim of at least 1 USDC, and *Addresses* counts the distinct holders, who can enter several groups. *Withdraw (%)* and *Deposit (%)* are the shares of owner-days with a completed withdrawal or deposit, and *Cash<Q* counts owner-days on which withdrawable cash is below the claim.

The treasury's gross withdrawals exceed a quarter of its prior claim on a few days, and on most of them large withdrawals are matched by deposits of similar size, so net outflows are

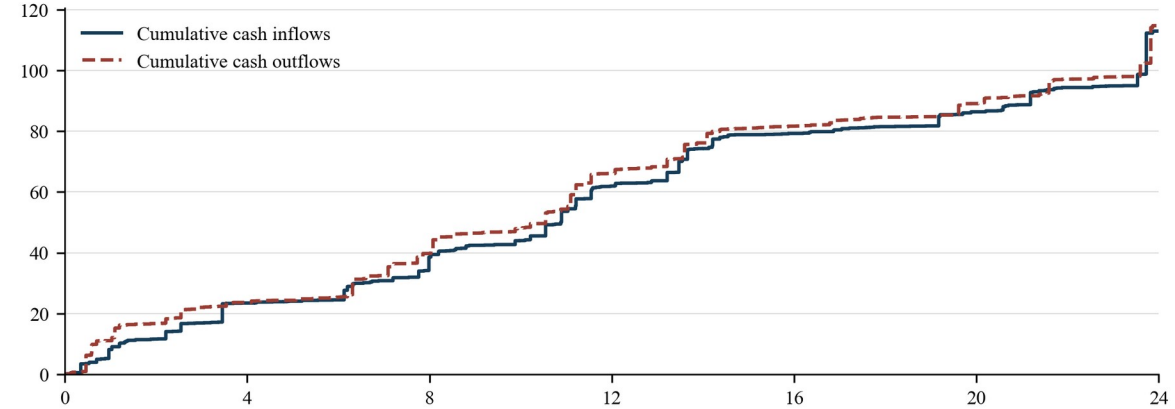
much smaller. Net withdrawals exceed a quarter of the prior claim only on 1 April and 8 April. On 8 April, the treasury withdraws more than the idle cash in the market at the start of the day, and Figure C2 shows that supplies and repayments by other participants arrive before each withdrawal, so total payments over a day can exceed the opening cash in the same market in which a single request for a quarter of the claim would fail at a given moment.

Figure C2. Cash replenishment on 8 April 2025

Panel A. Idle cash



Panel B. Cumulative flows



Notes. Both panels cover the cbBTC/USDC market of Morpho on Base on 8 April 2025, and the horizontal axis is the hour of the day in UTC. In Panel A, the vertical axis is idle cash in million USDC, stepped at every event, and circles mark the treasury's three withdrawals and their amounts. In Panel B, the vertical axis is cumulative cash into and out of the market since midnight, in million USDC. Inflows are supplies and repayments, and outflows are withdrawals and borrows.

Low coverage is a feature of very large claims and not of the typical holder. Panel B of Table C1 groups the vault's other holders by the size of their opening claim: of the eligible owner-days outside the treasury, only a few dozen have withdrawable cash below the full claim, none below one quarter of it, and all of them belong to Ethena. When we scale the vault's market positions and the treasury's claim down to a quarter of their size while holding market cash fixed, the treasury's median coverage would be several times higher, and a request for a quarter of the claim would exceed capacity on a single day instead of most days, so what makes the treasury's coverage low is its own size relative to cash that others share.

Availability of data and materials

All data come from public blockchain records of Ethereum, Base, Arbitrum and Polygon, read from the protocols' contracts. The code and the derived datasets that reproduce the tables and figures are available from the corresponding author on reasonable request.

CRediT authorship contribution statement

XinYu Zhang: Conceptualization, Data curation, Formal analysis, Writing – original draft. Hyoung-Goo Kang: Conceptualization, Supervision, Writing – review & editing. Han Sun: Methodology, Software, Validation, Writing – review & editing. Doojin Ryu: Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Claude (Anthropic) to edit the language and Codex (OpenAI) to write the statistical code. The authors reviewed and edited all output and take full responsibility for the content of the publication.

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